

EGERTON



UNIVERSITY

BMED 112: BIOCHEMISTRY OF BIOMOLECULES

By

Ms. Maiyo

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The course overview

This course will provide an introduction to biomolecules in living matter. The simplicity of the building blocks of macromolecules (amino acids, monosaccharides, fatty acids and purine and pyrimidine bases) will be contrasted with the enormous variety and adaptability that is obtained with the different macromolecules (proteins, carbohydrates, lipids and nucleic acids). The nature of the electronic and molecular structure of macromolecules and the role of non-covalent interactions in an aqueous environment will be highlighted. You are therefore expected to complete the course in 45 hours within a period of one semester.

The following topics will be addressed:

Topic One	Introduction to biomolecules Carbohydrates: Monosaccharides and their stereochemistry, Disaccharides, Polysaccharides and their biological differences
Topic Two	Nucleic acids: Nucleosides and heterocyclic chemistry, including tautomerisation, Nucleotides, Oligonucleotides and their synthesis, DNA/RNA structure
Topic Three	Peptides: amino acids and their organic chemistry, synthesis and structural properties of peptides, protein structure and function, protein-nucleic acid recognition
Topic Four	Lipids: structure, hydrophobic and hydrophilic properties, lipids and proteins in membranes

The course content

The course is broken down into topics. Each topic comprises:

- An introduction to the topic content
- New terminology

- Core content of the topic
- Assignments and/or assessments, as applicable

The unit will be delivered through lectures and related feedback sessions to ensure students fully understand what is expected of them. Short tests (formative assessment) will be used throughout the unit to test students' knowledge and monitor that the right material has been extracted from the lectures.

Course learning Outcomes

At the end of the course, the students will be expected to:



- A basic understanding of the composition, structure and function of the major groups of Monosaccharides and their stereochemistry, Disaccharides, Polysaccharides and their biological differences.
- A basic understanding of Nucleosides and heterocyclic chemistry, including tautomerisation, Nucleotides, Oligonucleotides and their synthesis, DNA/RNA structure
- Understand amino acids and their organic chemistry, synthesis and structural properties of peptides, protein structure and function, protein-nucleic acid recognition
- A basic understanding of lipids structure, hydrophobic and hydrophilic properties, lipids and proteins in membranes

Course Study Skills

All students undertaking this course in biochemistry and molecular biology should acquire a sound understanding of Biochemistry as well as:

- Become intellectually independent learners
- Recognize and confront areas of personal ignorance

- Review and apply chemical, biological, physical and mathematical principles in a biochemical context
- Improve problem-solving skills
- Create, understand and value abstract biochemical models
- See biochemistry in relevant historical and societal contexts
- Discover and use the resources of the library and the Internet
- Gain confidence in reading and understanding scientific articles
- Appreciate importance of clear oral and written communication

For this class, *as a group*, to become sufficiently skilled learners to score significantly well as they lay the foundation of the four years journey. In order for this to happen, you will need to work individually and together so that *everyone* learns. Everyone here has unique and special abilities, background and personality to contribute to the effort.

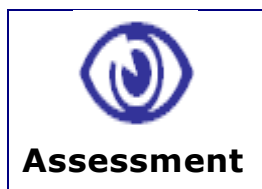
Course Assignments/Activities



Two assignments will be given. The assignments will be online or orally presented as a group activity on campus. They will be given at the beginning of suitable topics and submitted in 2 weeks' time. The assignment will make 5%

of the course.

Assessments



The course will comprise 2 continuous assessment tests marked out of 30 and converted to 25%. The main Exam will be marked out of 70. The CAT will be done during the 5th and 9th week of the semester. The assessments will be a 1

hour teacher-marked test. More communication on online assessment in the course of the semester as we deal with the pandemic.

Self-assessment

Self-assessments are provided in order to aid your understanding of the topic and course content. While they may not be graded, you are strongly advised to attempt them whenever they are available in a topic.

Need Help?

This course was developed in January 2021 by Ms. Maiyo, a lecturer in the Department of Biochemistry and Molecular Biology at Egerton University. Email address cmaiyo@gmail.com, Phone +254722805995.

This session, the instructor for this course is Ms. Maiyo. My office is located in the Science Complex, Room 302B located in the 2nd floor Biochemistry wing. You may consult me during the normal working hours between Tuesday and Thursday in the office or contact me through: **Phone or Email:**

For Further reading

Core text:

Nelson DL, Cox MM, Lehninger Principles of Biochemistry, 7th Edition, W.H. Freeman, 2017

Alternative core texts

Berg JM, Stryer L, Tymoczko JL, Gatto GJ, Biochemistry, 8th Edition, Macmillan HE, 2015

Garrett RH, Grisham CM, Biochemistry, 6th (international) Edition, Cengage, 2017

Background reading:

Alberts B, Essential Cell Biology, 4th Edition, W W Norton (ex Garland Press), 2014

Taylor MR, Simon EJ, Reece JB, Dickey J, Hogan KA, Campbell NA, Campbell Biology: Concepts & Connections, 9th Edition, Pearson, 2018

Recommended Reading:

Catch-Up Reading: Crowe J, Bradshaw T, Chemistry for the Biosciences: The Essential Concepts, 3rd Edition, Oxford University Press, 2014

TOPIC ONE

INTRODUCTION TO BIOMOLECULES

Introduction

Biochemistry is the study of the chemistry of cells and organisms. Thus it is concerned with the types of molecules found in biological systems, their structure and their chemical properties. Biochemistry also deals with the function of these molecules, how they interact and what reactions they undergo.

Properties of Biomolecules to be discussed include:

A. General Properties

Biomolecules are organic molecules, not fundamentally different from other, typical organic molecules. They are the same types of molecules, react in the same ways and obey the same physical laws.

B. Composition and Structure

Biomolecules contain mainly carbon, which behaves as it always does in organic compounds, forming 4 bonds, usually with a tetrahedral arrangement. The carbon skeleton can be linear, branched, cyclic or aromatic. Other important elements are H, O, N, P and S. About 30 elements are required by biological systems, including iodine and many metals, though most of these are needed in only trace amounts.

Biomolecules contain the same types of functional groups as do organic molecules, including hydroxyl groups, amino groups, carbonyl groups, carboxyl groups, etc. However, many biomolecules are polyfunctional, containing two or more different functional groups which can influence each other's reactivity.

Biomolecules tend to be larger than typical organic molecules. Small biomolecules have molecular weights over 100, while most biomolecules have molecular weights in the thousands, millions or even billions. Because of their large size, the majority of biomolecules have specific 3-dimensional shapes. The atoms of a biomolecule are arranged in space in a precise way and proper arrangement is usually needed for proper function. The 3-dimensional shape is maintained by numerous non-covalent bonds between atoms in the molecule. Because of the weak nature of most noncovalent bonds and because of interactions between the biomolecule and the solvent, the biomolecule's structure is flexible rather than static.

C. Stereochemistry

As is common with organic compounds, many biomolecules exhibit stereochemistry. When four different types of atoms or functional groups are bonded to one carbon atom, the carbon is stereogenic (or chiral or asymmetric) and the 1 compound can exist in two different isomeric forms that have different configurations in space. The two configurations are mirror images of each other and are not superimposable. When two compounds are mirror images of each other they are called enantiomers or optical isomers, a subclass of stereoisomers. Enantiomers usually have identical chemical properties and differ only in the way they rotate plane polarized light or interact with other chiral compounds. Most biomolecules have several or many asymmetric carbons and so may have many diastereomers, a subclass of stereoisomers that are non-mirror images and have different properties. Stereochemistry is important because biological systems usually use only one specific isomer of a given compound.

Types of Biomolecules

Biomolecules can be divided into several major classes and a few minor classes.

A. Amino Acids and Proteins

Amino acids are relatively small molecules with molecular weights around 100-200. They are used to produce energy, to synthesize other molecules like hormones and to make proteins. Proteins are polymers of amino acids. They fold into specific shapes and range in molecular weight from several thousand to over a million. Proteins function as enzymes (which catalyze reactions), structural elements, transport molecules, antibodies, etc.

B. Carbohydrates (sugars & starches)

The smallest carbohydrates are the monosaccharides with molecular weights of around 100-200. They are a major source of energy for biological systems. Polysaccharides are polymers of monosaccharides with molecular weights often in the millions. Polysaccharides also have definite shapes and serve as structural elements or as stored metabolic energy.

C. Lipids (fats & oils)

Lipids are relatively small water-insoluble molecules with molecular weights of up to 750-1500. Because they are defined by their water-insolubility, they are chemically more diverse than the other classes of biomolecules, with about half a dozen major types. Lipids are used for energy production and storage, hormones, structural elements of cell membranes, and vitamins. Lipids do not polymerize to form macromolecules, but they can aggregate non-covalently to form very large structures.

D. Nucleotides and Nucleic Acids

Nucleotides are relatively small molecules with molecular weights in the hundreds. They function in transferring energy and in helping enzymes to catalyze reactions. Nucleic acids (DNA and RNA) are large polymers of nucleotides, with molecular weights up into the billions. They form structures

like the double helix and they function in storing, transmitting and utilizing genetic information.

E. Small Organic Molecules

In addition to the major classes of biomolecules, there are many relatively small organic molecules required by cells for very specific functions; these molecules do not fall neatly into one of the above major categories. These molecules can be precursors of biomolecules that help enzymes function (often related to vitamins) or can be intermediates in metabolic pathways, etc.

F. Inorganic Ions

Though not actually biomolecules, many inorganic ions are required by cells, often in trace amounts. These include calcium, sodium, iron, magnesium, potassium, chlorine, etc. Inorganic ions perform a variety of functions such as structural elements (calcium in bone), regulation of osmotic pressure and transport (sodium), and components of proteins and enzymes (iron).

G. Combinations of Biomolecules

Sometimes one biomolecule can contain components from two of the major classes, such as a lipoprotein (lipid plus protein) or a glycoprotein (carbohydrate plus protein)

TOPIC TWO

CARBOHYDRATES

BIOMEDICAL IMPORTANCE

Carbohydrates are widely distributed in plants and animals; they have important structural and metabolic roles. In plants, glucose is synthesized from carbon dioxide and water by photosynthesis and stored as starch or used to synthesize the cellulose of the plant cell walls. Animals can synthesize carbohydrates from amino acids, but most are derived ultimately from plants.

Glucose is the most important carbohydrate; most dietary carbohydrate is absorbed into the bloodstream as glucose formed by hydrolysis of dietary starch and disaccharides, and other sugars are converted to glucose in the liver.

Glucose is the major metabolic fuel of mammals (except ruminants) and a universal fuel of the foetus. It is the precursor for synthesis of all the other carbohydrates in the body, including **glycogen** for storage; **ribose** and **deoxyribose** in nucleic acids; **galactose** for synthesis of lactose in milk, in glycolipids and in combination with protein in glycoproteins and proteoglycans. Diseases associated with carbohydrate metabolism include **diabetes mellitus**, **galactosemia**, **glycogen storage diseases** and **lactose intolerance**

CARBOHYDRATES ARE ALDEHYDE OR KETONE DERIVATIVES OF POLYHYDRIC ALCOHOLS

Carbohydrates are classified as follows depending on size:

1. **Monosaccharides** are those sugars that cannot be hydrolyzed into simpler carbohydrates. Examples are listed in the **Table below**.

	Aldoses	Ketoses
Trioses ($C_3H_6O_3$)	Glycerose (glyceraldehyde)	Dihydroxyacetone
Tetroses ($C_4H_8O_4$)	Erythrose	Erythrulose
Pentoses ($C_5H_{10}O_5$)	Ribose	Ribulose
Hexoses ($C_6H_{12}O_6$)	Glucose	Fructose
Heptoses ($C_7H_{14}O_7$)	—	Sedoheptulose

They are white solids, water-soluble (polar) and often have a sweet taste. Their chemical formula is $(CH_2O)_n$ where $n = 3-7$. The most common number of carbons is 5 or 6. The carbon skeleton is unbranched, connected by single bonds. One carbon contains a carbonyl oxygen (aldehyde or ketone). All other carbons contain a hydroxyl group. Monosaccharides are named with the suffix -ose. Every monosaccharide can be classified as an aldose or ketose, depending upon the functional group. They can also be classified according to the number of carbon atoms: 3 carbons = **a triose**; 4 C = **a tetrose**; 5 C = **a pentose**; 6 C = **a hexose**; 7 C = **a heptose**.

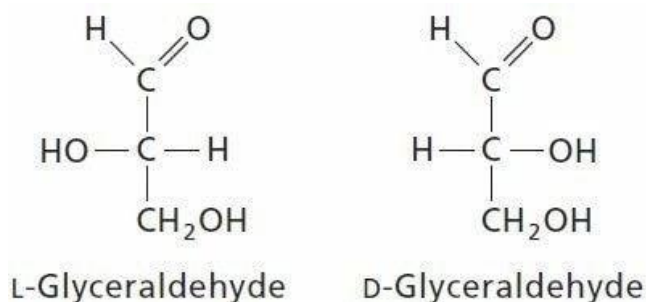
The two classifications are often combined, such as aldopentose or ketohexose.

Structure and Stereochemistry

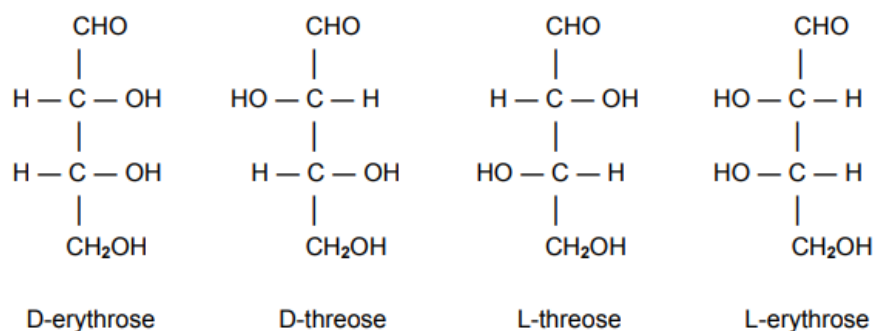
The most important classification is aldose vs. ketose.

A: Aldoses-The simplest aldose contains three carbons with one aldehyde and two hydroxyl groups. The name for this compound is glyceraldehyde. Glyceraldehyde contains a chiral or asymmetric carbon, also referred to as a stereogenic or chirality centre. The central carbon is attached to four different substituents, giving two forms that are mirror images of each other but are not superimposable. They are designated D and L. All other monosaccharides

are classified as D or L, depending on the configuration of the asymmetric carbon farthest from the aldehyde group.

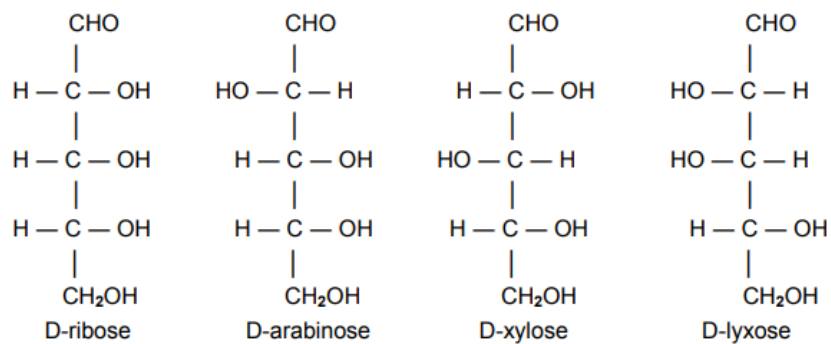


A molecule with n chiral centers will have 2^n stereoisomers. Thus 4-carbon aldoses have two stereogenic centres and four stereoisomers.

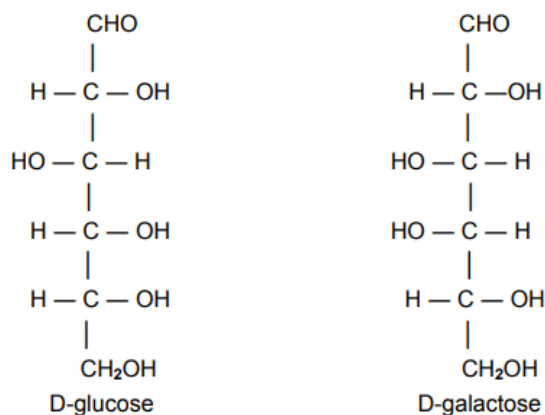


D-erythrose D-threose L-threose L-erythrose

The first two structures are D-sugars because the configuration around the third carbon is like that of D-glyceraldehyde. The last two structures are L-sugars because they resemble L-glyceraldehyde. D-threose and L-threose are mirror images (enantiomers) and will have all the same properties except their behavior with plane polarized light and their interaction with other chiral compounds. D-erythrose and D-threose are not mirror images (are diastereomers) and are given **different names because they are different compounds with different properties**. The D-sugars are the biologically important ones. There are four D-aldopentoses and eight D-aldohexoses (and the same number of L-isomers).



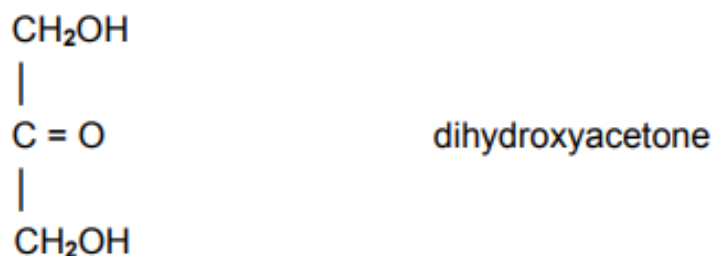
The most important aldoses and the most common in biological systems are D-ribose (an aldopentose) and two aldohexoses, D-glucose and D-galactose.



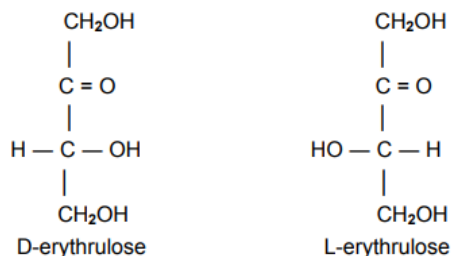
Sugars that differ in configuration around one carbon are called epimers. D-glucose and D-galactose are C-4 epimers.

B: Ketoses

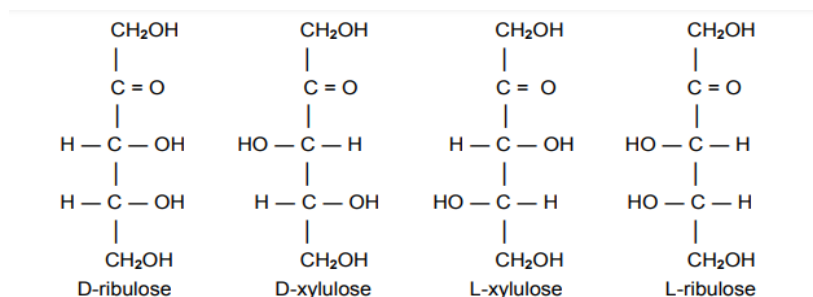
The simplest ketose will have three carbons, one ketone group and two hydroxyl groups.



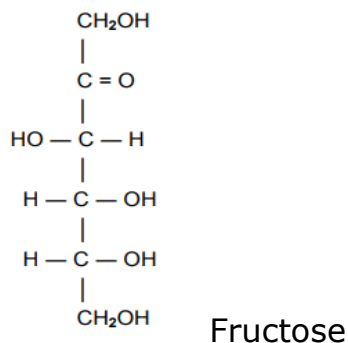
This compound does not have a chiral carbon and so does not have D and L forms. When a fourth carbon is added, a stereogenic centre is created.



D and L designations are based on glyceraldehyde. If the OH on the last chiral carbon points right, it is a D isomer. If the OH points left it is an L-isomer. For ketopentoses, there are two chiral carbons and four stereoisomers.

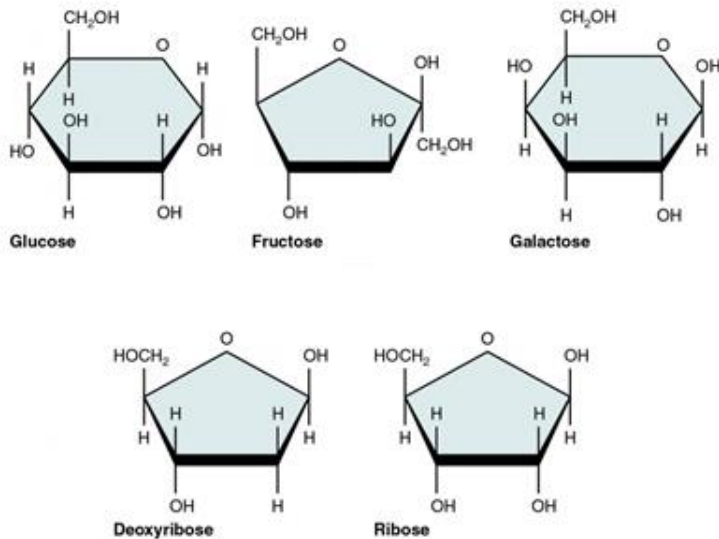


The various stereoisomers again form pairs of enantiomers or diastereomers. The D-isomers are the biologically important ones. Ketoses are generally named by inserting 'ul' into the name of the corresponding aldose (ribose-ribulose).

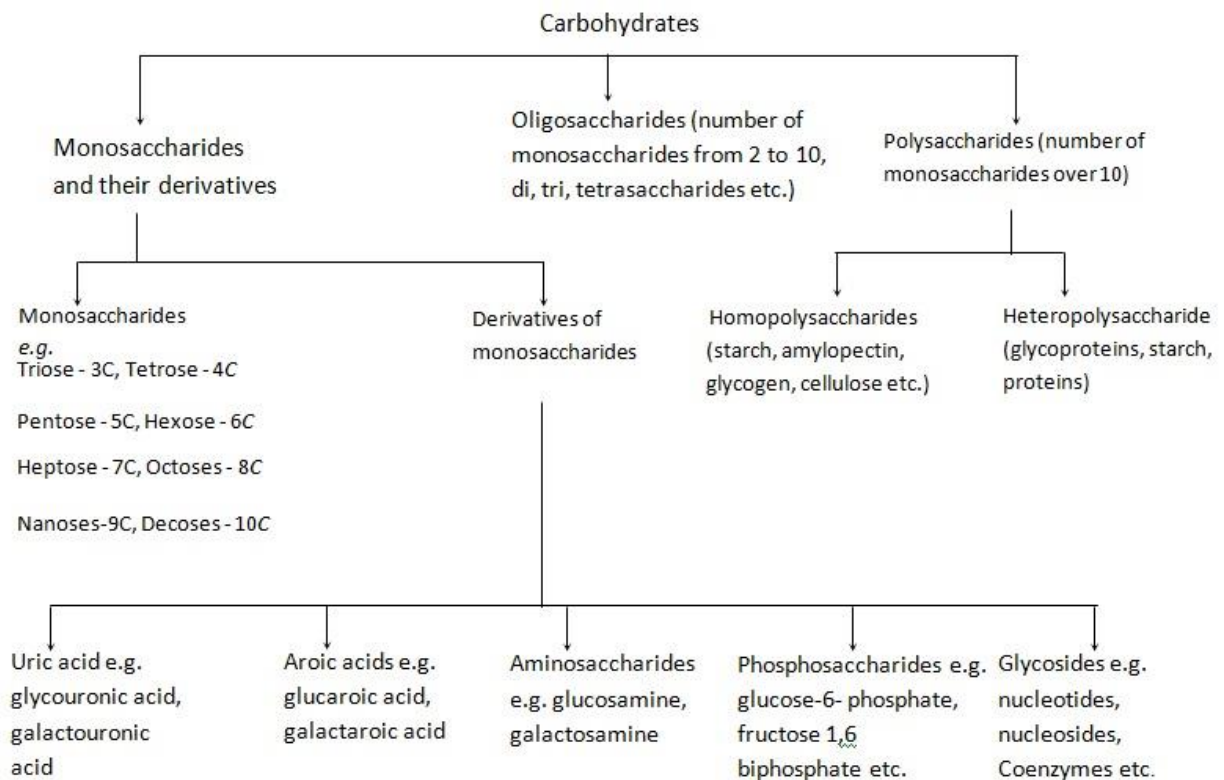


The most important ketose is D-fructose, which is a ketohexose.

When there are five or six carbons present, other ketoses are possible where the ketone group is on carbon 3. However, all the biologically important ketoses have the ketone group on carbon 2.



Important Monosaccharides

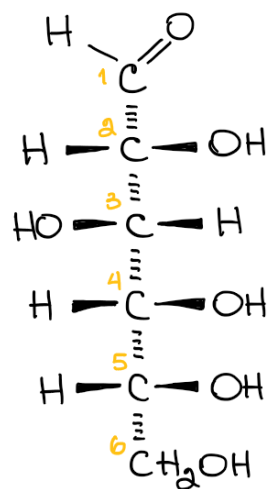


BIOMEDICALLY, GLUCOSE IS THE MOST IMPORTANT MONOSACCHARIDE

The Structure of Glucose Can Be Represented in Three Ways

The straight-chain structural formula (aldohexoses) can account for some of the properties of glucose, but a cyclic structure (a **hemiacetal** formed by reaction between the aldehyde group and a hydroxyl group) is thermodynamically favoured and accounts for other properties. The cyclic structure is normally drawn as shown in the Haworth projection, in which the molecule is viewed from the side and above the plane of the ring; the bonds nearest to the viewer are bold and thickened, and the hydroxyl groups are above or below the plane of the ring. The six-membered ring containing one oxygen atom is actually in the form of a chair projection.

Fischer Projections 1

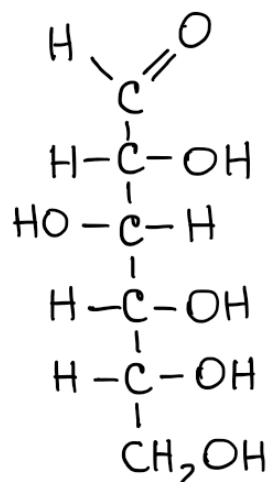


D-glucose

Fischer projection of glucose with stereochemistry shown

As you probably know, Fischer projections are a very convenient way of drawing a long-chain molecule with a lot of chiral carbons, such as a carbohydrate. In a Fischer projection, we put the most oxidized atom (an

aldehyde function) on the top and the rest of the chain following from there. For example, look at the structure of glucose drawn out in the Fischer Projection form 2.



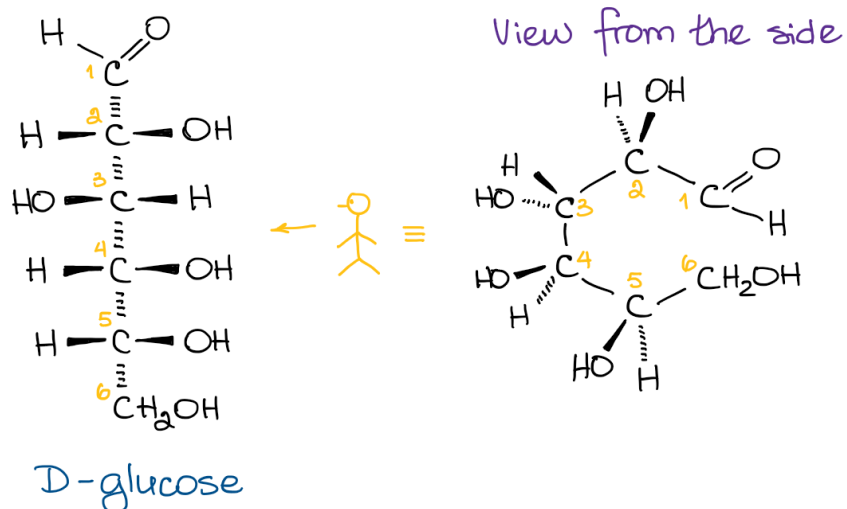
D-glucose

Fischer projection of glucose without stereochemistry shown

On the projection 1, you have a glucose structure in the Fischer projection with the stereochemistry shown. Every vertical line in the Fischer projection is oriented away from the observer, every horizontal line looks toward the observer. Since this is a common stereochemical convention, we don't need to show it every single time for the simplicity sake.

What's important though, is to remember this convention at all times. This means, that Fischer projections CANNOT be flipped in space around their vertical axis as this would make an enantiomer of the molecule! So while we might be drawing it "flat" it actually is not and needs to be treated like a 3D object. When it comes to the 3D representation of the Fischer projections, I really like the image of a caterpillar standing on its hind legs, bending backwards, while trying to hug you. Let me illustrate what I'm trying to describe here:

I'm assuming you're somewhat familiar with the Fischer projections and this all is just a refresher. So, let's look at how you can convert the Fischer projection into Haworth projection.

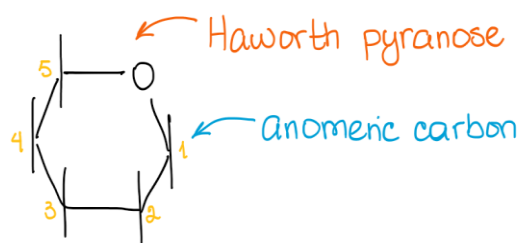


Fischer projections of glucose, view from the side

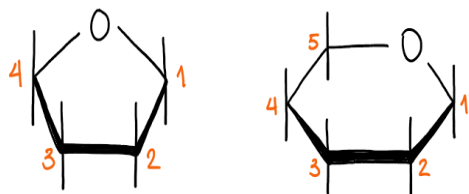
Haworth Projections

Here will be focusing on the pyranoses. Pyranose is a 6-membered ring that is traditionally drawn with the anomeric carbon to the right:

Haworth Projection (pyranose)



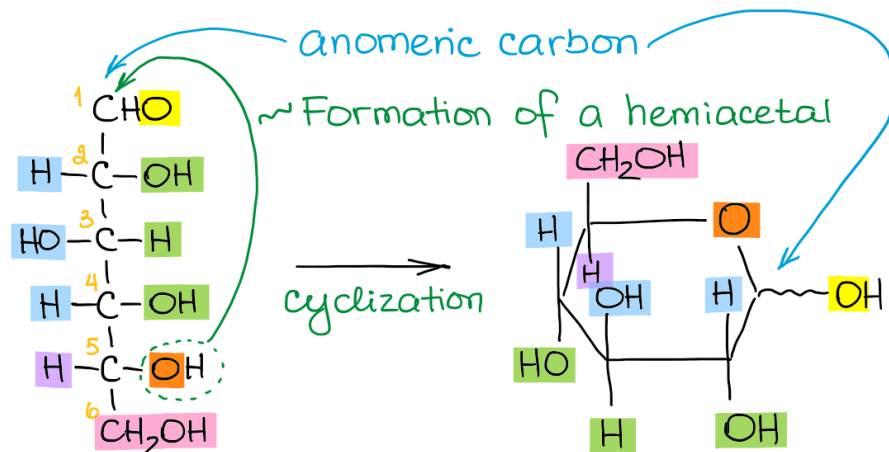
Haworth projection of a Pyranose



The "bottom" portion of the Haworth projection is oriented towards the observer

projection

Stereochemistry of a Haworth



D-glucose

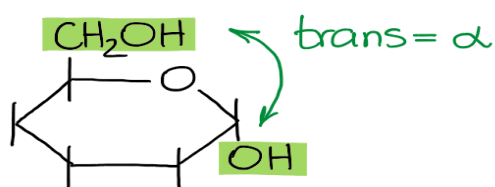
D-glucopyranose

Conversion of Fischer

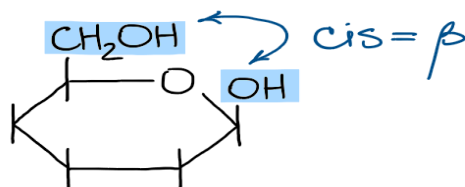
projection into Haworth projection

The Stereochemistry of the Anomeric Carbon (the α -form or the β -form)

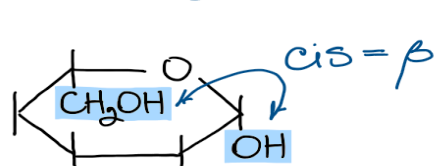
The anomeric carbon has a special place in the chemistry of carbohydrates because it doesn't have a set stereochemistry. This means that it can be either facing "up" or "down" depending on how the cyclization happened. While, there are factors that can make one orientation more favorable than the other one.



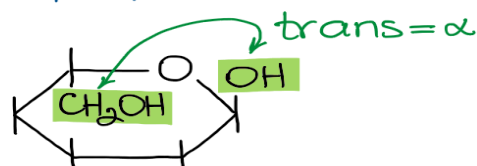
D- α -pyranose



D- β -pyranose



L- β -pyranose

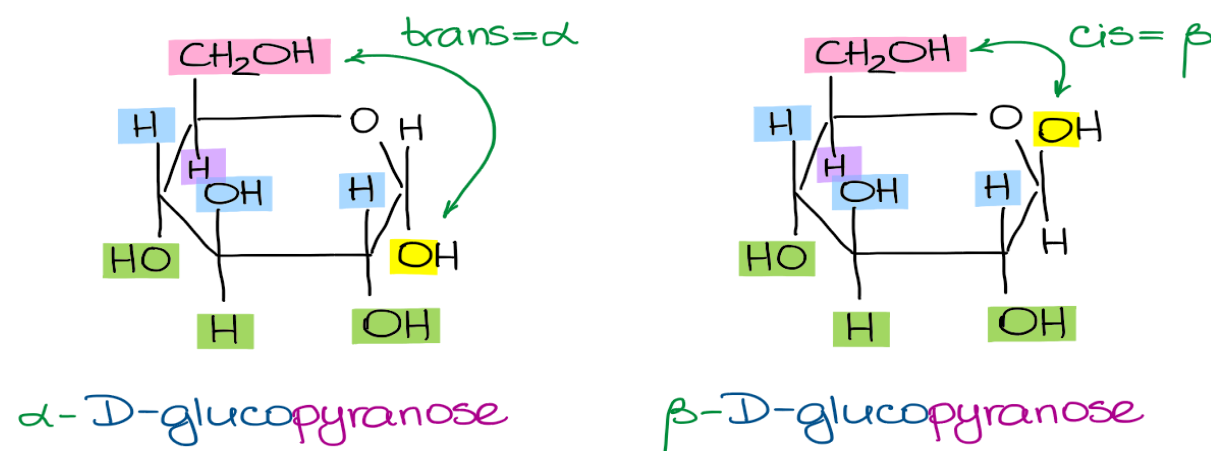


L- α -pyranose

The α - and the β -forms of pyranoses

The stereochemistry of the anomeric carbon can be either the α -form or the β -form. By the definition, for a pyranose saccharide, the α -form is the one where the anomeric -OH and the carbon #6 are trans to each other, while the β -form has those two groups in a cis configuration.

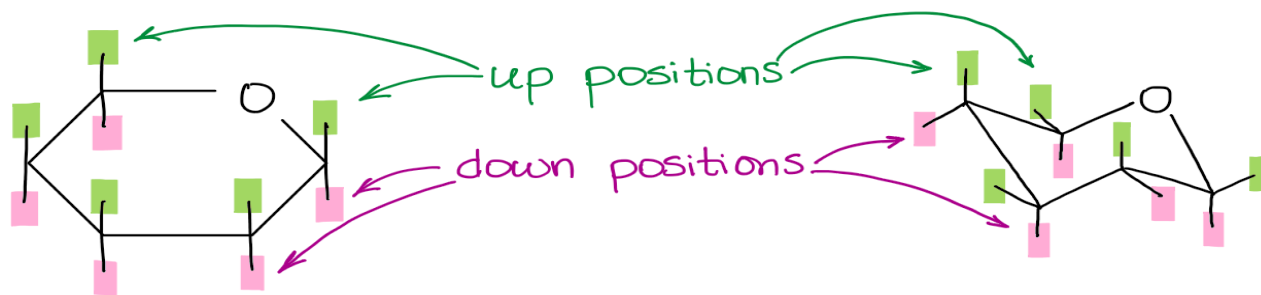
Thus, I can draw either α - or the β -forms of any of my molecules from above. For instance, here are the α -D-glucopyranose and the β -D-glucopyranose:



The α -D-glucopyranose and the β -D-glucopyranose

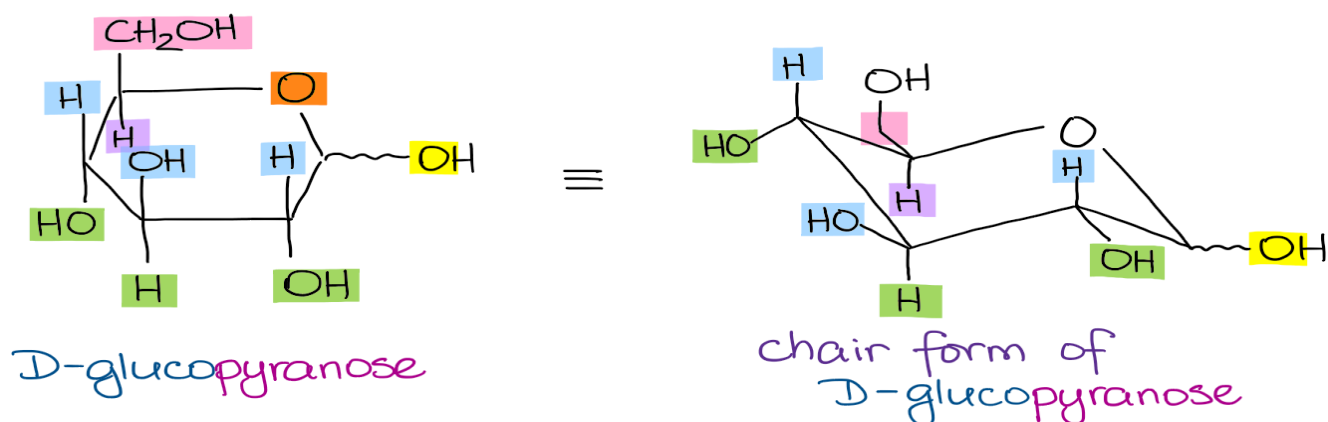
Converting Haworth to Chair

The final piece of these conversions is often to draw a complete chair conformation of the pyranose. The trick is to remember that just like the Haworth projections; the chair conformations also have the well-defined “up” and “down” positions!



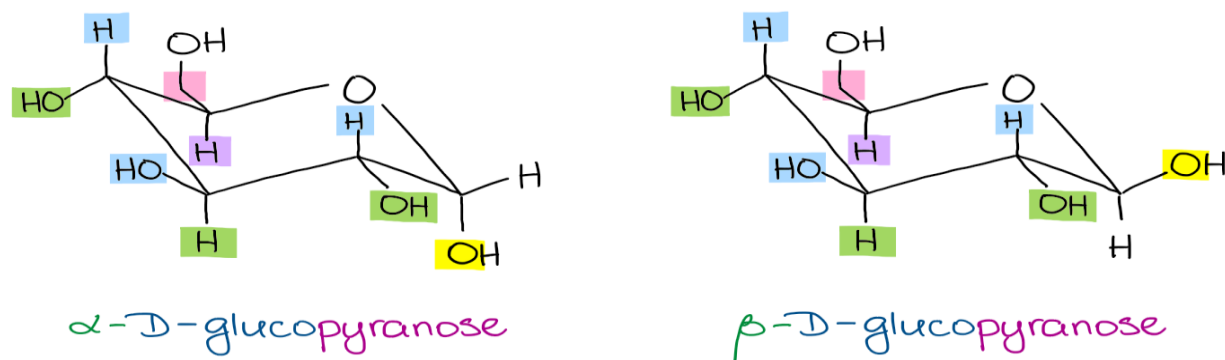
Up and down positions in the Haworth projection and a chair conformation

Looking at the glucopyranose from above, you now can easily draw the chair conformation for it in a general form:



Haworth projection and chair conformation of glucopyranose

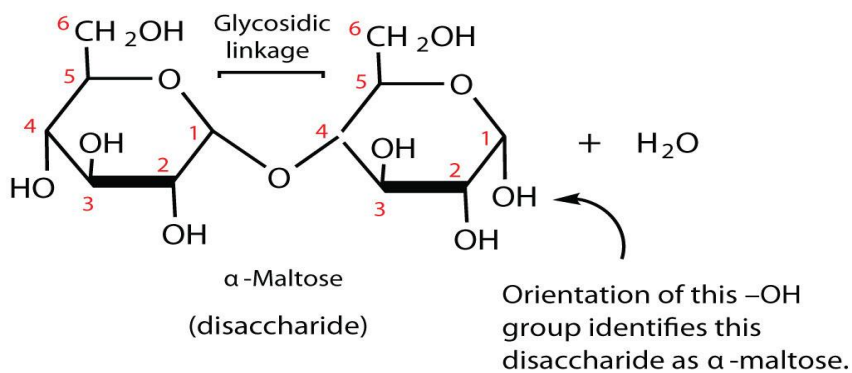
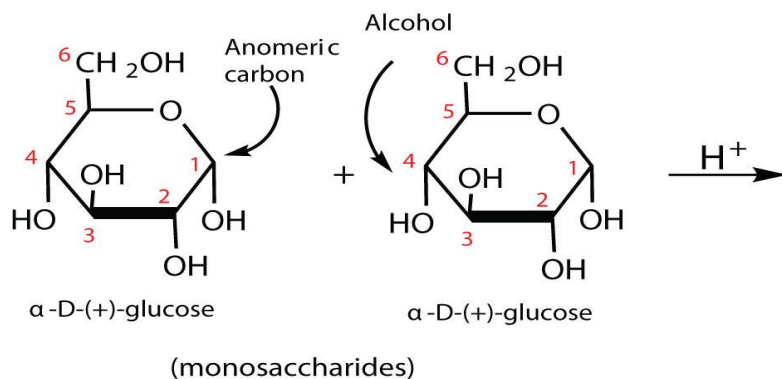
And, just as easily, you can show chair conformations specifically for the α -D-glucopyranose and the β -D-glucopyranose:



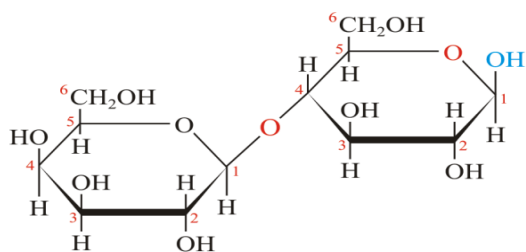
Chair conformations of alpha and beta D-glucopyranose

2. Disaccharides are condensation products of two monosaccharide units, for example, lactose, maltose, sucrose and trehalose. The bond that holds the monosaccharides together is called a **glycosidic bond**. The bond is stable at neutral and basic pH but is broken by acid hydrolysis. There are two basic characteristics to any disaccharide. The first is the identity of the monosaccharide components. These may be the same or different. The second is the nature of the glycosidic bond. The bond is characterized by which

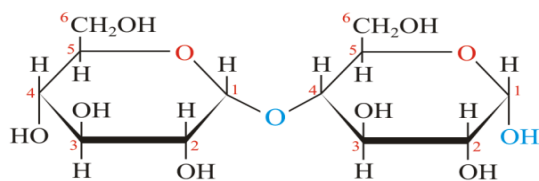
carbons of the monosaccharides are linked and by the orientation of the bond (α or β).



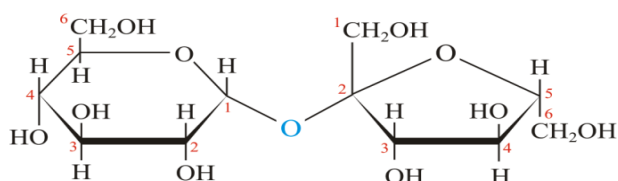
β -lactose
 β -D-galactopyranosyl-(1 $\rightarrow$ 4)- β -D-glucopyranose



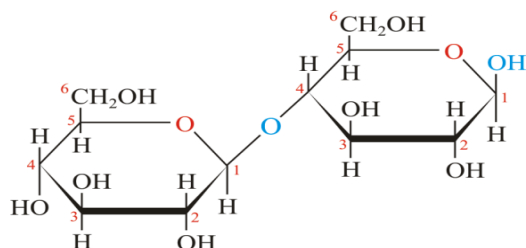
α -maltose
 α -D-glucopyranosyl-(1 $\rightarrow$ 4)- α -D-glucopyranose



sucrose
 α -D-glucopyranosyl-(1 $\rightarrow$ 2)- β -D-fructofuranoside



β -cellobiose
 β -D-glucopyranosyl-(1 $\rightarrow$ 4)- β -D-glucopyranose



List of disaccharides with their monomer units and functions

Disaccharide	First Unit	Second Unit	Bond	Function
Sucrose	Glucose	Fructose	$\alpha(1\rightarrow2)\beta$	It's a product of photosynthesis
Lactulose	Galactose	Fructose	$\beta(1\rightarrow4)$	Non-absorbable sugar used in the treatment of constipation and hepatic encephalopathy
Lactose	Galactose	Glucose	$\beta(1\rightarrow4)$	A major animal energy source
Maltose	Glucose	Glucose	$\alpha(1\rightarrow4)$	Important intermediate in starch and glycogen digestion
Trehalose	Glucose	Glucose	$\alpha(1\rightarrow1)\alpha$	Energy source for insects
Cellobiose	Glucose	Glucose	$\beta(1\rightarrow4)$	Essential in carbohydrate metabolism

3. Oligosaccharides are condensation products of three to ten monosaccharides. Most are not digested by human enzymes. They are used as prebiotics.

Prebiotics are non-digestible nutrients that selectively promote the growth of normal intestinal bacteria that may have beneficial effects on the large intestinal lining. Oligosaccharides currently considered as prebiotics include fructooligosaccharides (FOS) or oligofructose and trans-galactooligosaccharides (TOS)

Note: Galacto-oligosaccharides (GOS):= **raffinose**, **Stachyose** and **verbascose** (in beans, peas, lentils, cabbage, whole grains), soybean oligosaccharides in soy

Trisaccharides are oligosaccharides comprised of three monosaccharides.
Examples include:

- Nigerotriose 3 glucose units joined by $\alpha(1\rightarrow3)$ glycosidic linkage,
- Maltotriose 3 glucose units joined by $(1\rightarrow4)$ glycosidic linkage
- Melezitose (glucose-fructose-glucose)
- Maltotriulose (glucose-glucose-fructose)
- Raffinose (galactose-glucose-fructose) and
- Kestose (glucose-fructose-fructose)

Tetrasaccharides are oligosaccharides comprised of four monosaccharides.

Examples are:

- Nigerotetraose 4 glucose units joined by $\alpha(1\rightarrow3)$ glycosidic linkage,
- Maltotetraose 4 glucose units joined by $(1\rightarrow4)$ glycosidic linkage
- Lychnose (galactose-glucose-fructose-galactose)
- Nystose (glucose-fructose-fructose-fructose)
- Sesamose (galactose-galactose-fructose-glucose)
- Stachyose (galactose-galactose-glucose-fructose)

Pentasaccharides are those comprised of five sugar units. N-linked oligosaccharides are mostly pentasaccharide. Examples of physiologically important pentoses

Hexasaccharides are oligosaccharides comprised of six sugar units. α -Cyclodextrin is an example. It consists of six glucose units linked via α -1, 4 linkages. Examples

Heptasaccharides are oligosaccharides containing seven sugar units

Octasaccharides-contain eight,

Nonasaccharides-have nine

Decasaccharides-have ten

4. Polysaccharides are condensation products of more than ten monosaccharide units; examples are the starches and dextrans, which may be

linear or branched polymers. Polysaccharides are sometimes classified as hexosans or pentosans, depending on the identity of the constituent monosaccharides (hexoses and pentoses, respectively). In addition to starches and dextrans, foods contain a wide variety of other polysaccharides that are collectively known as non-starch polysaccharides; they are not digested by human enzymes and are the major component of dietary fibre.

Examples are cellulose from plant cell walls (a glucose polymer) and inulin, the storage carbohydrate in some plants (a fructose polymer).

Special enzymes bind these small monomers together creating large sugar polymers, or polysaccharides. A polysaccharide is also called a **glycan**. A polysaccharide can be a **homopolysaccharide**, in which all the monosaccharides are the same, or a **heteropolysaccharide** in which the monosaccharides vary. Depending on which monosaccharides are connected, and which carbons in the monosaccharides connect, polysaccharides take on a variety of forms. A molecule with a straight chain of monosaccharides is called a linear polysaccharide, while a chain that has arms and turns are known as a branched polysaccharide.

Note: Polysaccharides do not have definite molecular weights since enzymes easily add or remove monosaccharide units.

Functions of a Polysaccharide

Depending on their structure, polysaccharides can have a wide variety of functions in nature. Some polysaccharides are used for storing energy, some for sending cellular messages and others for providing support to cells and tissues.

a) Storage of Energy

Many polysaccharides are used to store energy in organisms. While the enzymes that produce energy only work on the monosaccharides stored in a polysaccharide, polysaccharides typically fold together and can contain many

monosaccharides in a dense area. Further, as the side chains of the monosaccharides form as many hydrogen bonds as possible with themselves, water cannot intrude the molecules, making them **hydrophobic**. This property allows the molecules to stay together and not dissolve into the cytosol. This lowers the sugar concentration in a cell and more sugar can then be taken in. Not only do polysaccharides store the energy, but they allow for changes in the concentration gradient, which can influence cellular uptake of nutrients and water.

b) Cellular Communication

Many polysaccharides become **glycoconjugates** when they become covalently bonded to proteins or lipids. Glycolipids and glycoproteins can be used to send signals between and within cells. Proteins headed for a specific organelle may be “tagged” by certain polysaccharides that help the cell move it to a specific organelle. The polysaccharides can be identified by special proteins, which then help bind the protein, vesicle, or other substance to a microtubule. The system of microtubules and associated proteins within cells can take any substance to its destined location once tagged by specific polysaccharides. Further, multi-cellular organisms have immune systems driven by the recognition of glycoproteins on the surface of cells. The cells of a single organism will produce specific polysaccharides to adorn its cells with. When the immune system recognizes other polysaccharides and different glycoproteins, it is set into action, and destroys the invading cells.

c) Cellular Support

By far one of the largest roles of polysaccharides is that of support. All plants on Earth are supported, in part, by the polysaccharide **cellulose**. Other organisms, like insects and fungi, use **chitin** to support the extracellular matrix around their cells. A polysaccharide can be mixed with any number of other components to create tissues that are more rigid, less rigid or even materials with special properties. Between chitin and cellulose, both

polysaccharides made of glucose monosaccharides, hundreds of billions of tons are created by living organisms every year. Everything from the wood in trees, to the shells of sea creatures is produced by some form of polysaccharide. Simply by rearranging the structure, polysaccharides can go from storage molecules to much stronger fibrous molecules. The ring structure of most monosaccharides aids this process.

Structure of a Polysaccharide

All polysaccharides are formed by the same basic process: monosaccharides are connected via **glycosidic bonds**. When in a polysaccharide, individual monosaccharides are known as **residues**.

- The structure of the molecules being combined determines the structures and properties of the resulting polysaccharide. The complex interaction between their hydroxyl groups (OH), other side groups, the configurations of the molecules and the enzymes involved all affect the resulting polysaccharide produced.
- A polysaccharide used for energy storage will give easy access to the monosaccharides, while maintaining a compact structure.
- A polysaccharide used for support is usually assembled as a long chain of monosaccharides, which acts as a fibre. Many fibres together produce hydrogen bonds between fibres that strengthen the overall structure of the material.

Examples of a Polysaccharide

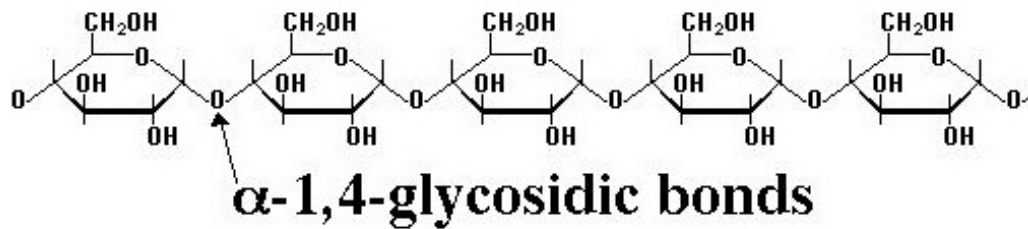
a) Starch

Starch is the storage polysaccharide of plants, occurring as granules inside cells, heavily hydrated with water. Starch has two components.

1. **α -Amylose** makes up ~20% of starch. It contains only D-glucose in an unbranched chain with the units linked by α 1,4 glycosidic bonds. Since virtually all the anomeric carbons are in glycosidic bonds, amylose is

nonreducing. It can be hydrolyzed by α -amylase to yield glucose and maltose. β -amylase hydrolyzes alternate bonds to produce maltose.

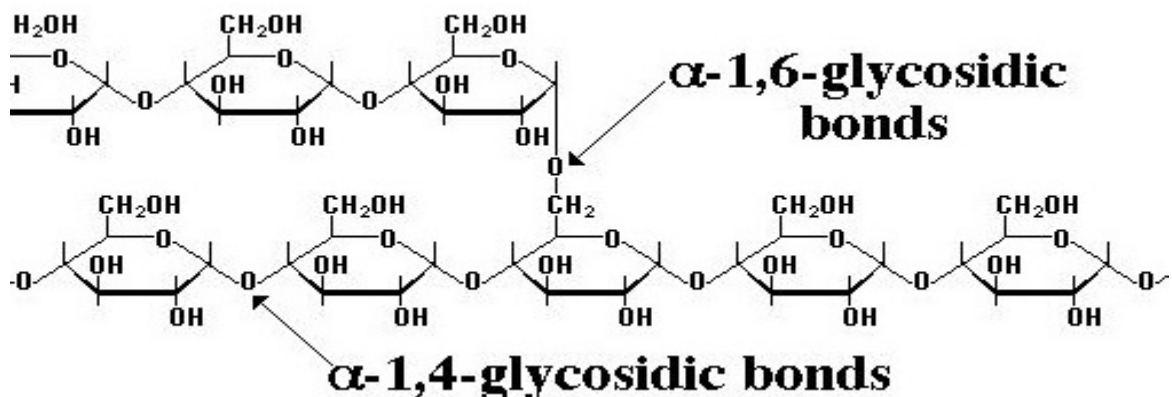
- It is water-soluble.
- It has 60-300 glucose units.
- It forms by C_1-C_4 glycosidic linkage. It has α 1-4 glycosidic linkages.



- Soluble in hot water but does not form a starch gel or paste. It does not form a gel when hot water is added.

2. Amylopectin

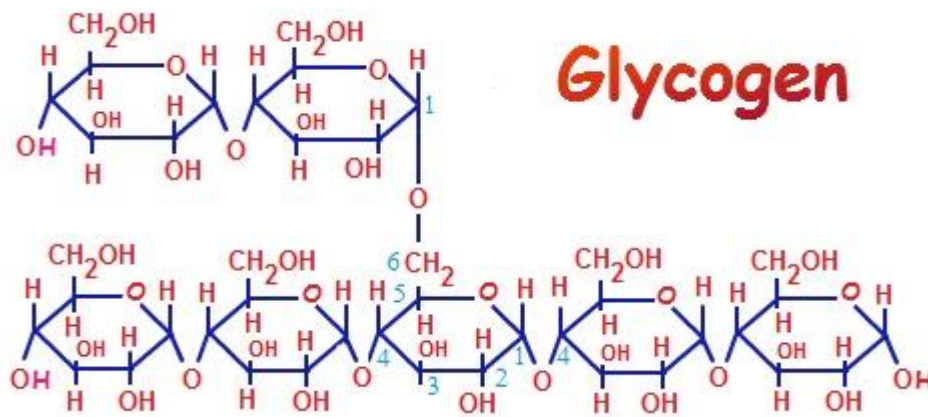
- It is a branched-chain polymer of α -glucose. It is a polysaccharide that is composed of monosaccharides. Monosaccharides are D-glucose molecules. Starch contains about 80% of amylopectin.
- It is water-insoluble.
- It is a branched structure.
- It constitutes about 90-80% of the starch.
- It has 2000-200,000 units of glucose.
- It forms by C_1-C_6 glycosidic linkage. It has α 1-4 glycosidic linkages and α 1-6 glycosidic linkages.



- Soluble in hot water with swelling. It forms a gel when hot water is added.
- Hydrolysis with Enzymes – Amylopectin cannot be hydrolyzed with α amylase and β amylase enzymes completely.

C. Glycogen

Glycogen is the storage polysaccharide of animals. It is very similar to amylopectin, but more highly branched (every 8-12 residues) and more compact.

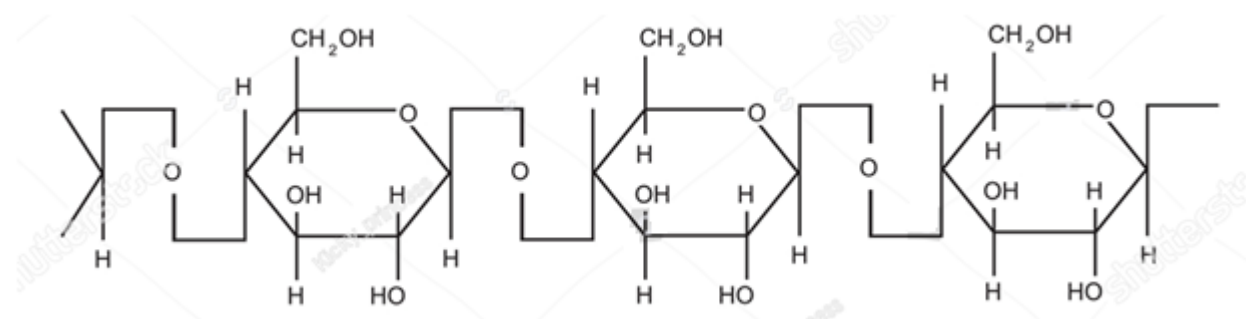


It is stored mainly in liver and skeletal muscle. Because it is branched, there are many sites at which enzymes can degrade it, allowing glucose to be released quickly when energy is needed. **Glucose cannot be stored in its monomeric form because it would raise the osmolarity of the cell too high, and create such a concentration gradient (high concentration in the cell) that more glucose could not be taken up.** Since glycogen is essentially insoluble, it does not cause similar problems.

D. Cellulose

Cellulose is found in the cell walls of plants where it provides support and structure. It consists entirely of D-glucose in linear chains of 10,000-15,000

monomers. However, cellulose contains β 1,4 linkages which make it quite different from the earlier polysaccharides.



Polysaccharide structure depends upon the type of covalent glycosidic bonds. That structure then tends to be stabilized by H-bonds between OH groups. When the bonds are α , the shape of the molecule is curved into a coil, which is a good compact shape for storage. This tight coil is stabilized by H-bonds. When the bonds are β , such as in cellulose the chain tends to be extended and linear, again stabilized by H-bonds. Several cellulose chains lying next to each other can form a network of bonds, resulting in straight, stable, strong fibers. Cellulose can be cleaved to glucose by the enzyme cellulase.

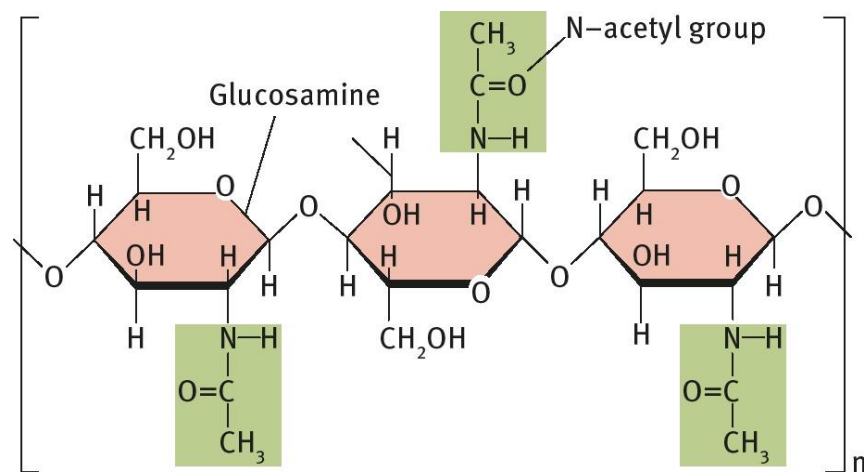
E. Chitin

Chitin forms the hard exoskeletons of arthropods (insects, lobsters, etc.). It is a homopolysaccharide of N-acetyl-D-glucosamine joined by β 1,4 linkages. Thus it will form strong, extended fibres. Although chitin is the dominant constituent, other compounds such as proteins and calcium carbonate also play a crucial role in the formation of exoskeleton.

Function

- This skeleton on the outside of the body appears hard and rigid due to the presence of chitin that is known for its tough elastic properties.
- The main function of this chitin-containing exoskeleton is to keep the inner soft tissue safe from any sort of injury.

- Most importantly, it prevents these delicate tissues from becoming dry. In short, it acts as a watertight barrier against dehydration, which is crucial for their survival.
- The hard chitin-containing exoskeleton of arthropods also acts as a defense mechanism against predation. This outer covering can tolerate strong compressive stresses, which can provide protection from predation because predators exert a compressive force on the exoskeleton to injure their victim.
- The fungal cell wall that protects the micro-organism from the outside environment is also made up of chitin.

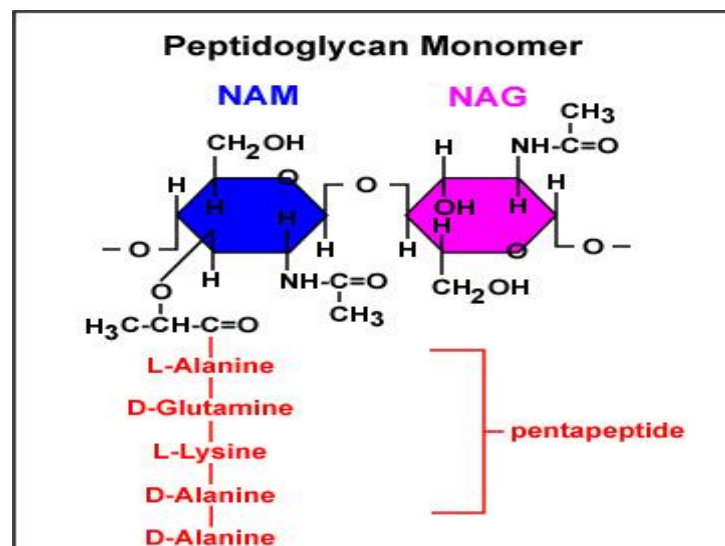


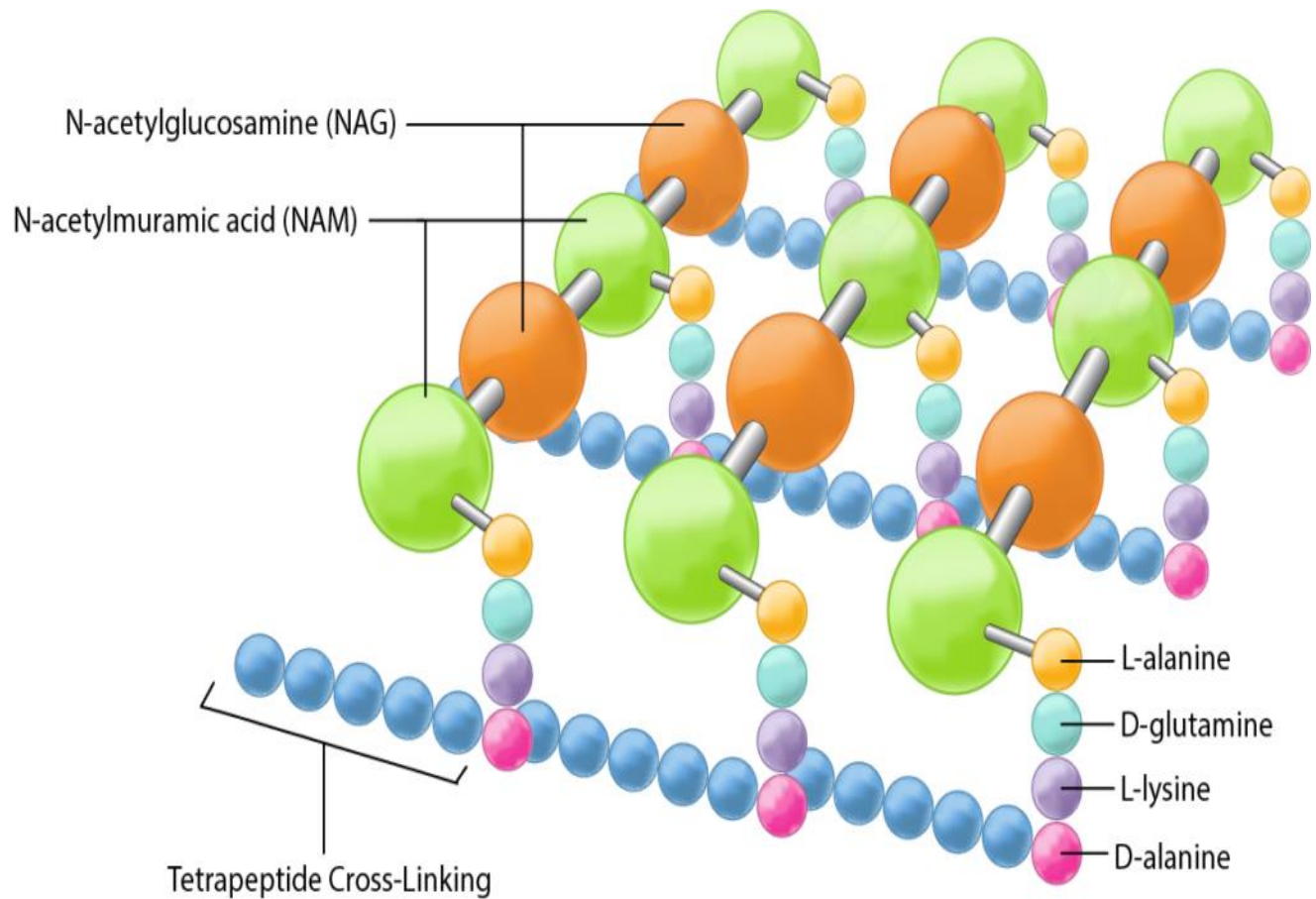
- Chitin is also used for manufacturing strong and flexible surgical threads. Quite a few dissolvable stitches used to close wounds are made from chitin.
- Use of food chitin can also help in creating stable food emulsions. It essentially acts as an excellent emulsifying agent, which helps to prevent the breaking of emulsion when exposed to other fluids. For instance, whipped dessert toppings often contain chitin that provides uniformity and stability to the product.
- Chitin has a long history of use as a food additive.
- One of the most important benefits of chitin is its use in making fertilizers. Chitin-containing fertilizers are organic, non-toxic and have

shown to increase crop productivity. Chitin in fertilizers helps in increasing soil organisms and enzyme activities, which positively affects soil health. This in turn increases crop yield.

F. Bacterial Cell Walls

The bacteria cell wall contains long parallel polysaccharide chains linked together by short peptides to form one enormous cage-like molecule called a **peptidoglycan**. The polysaccharide chains are heteropolymers of alternating N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM) linked by β 1,4 bonds. The exact nature of the peptide varies with the species.



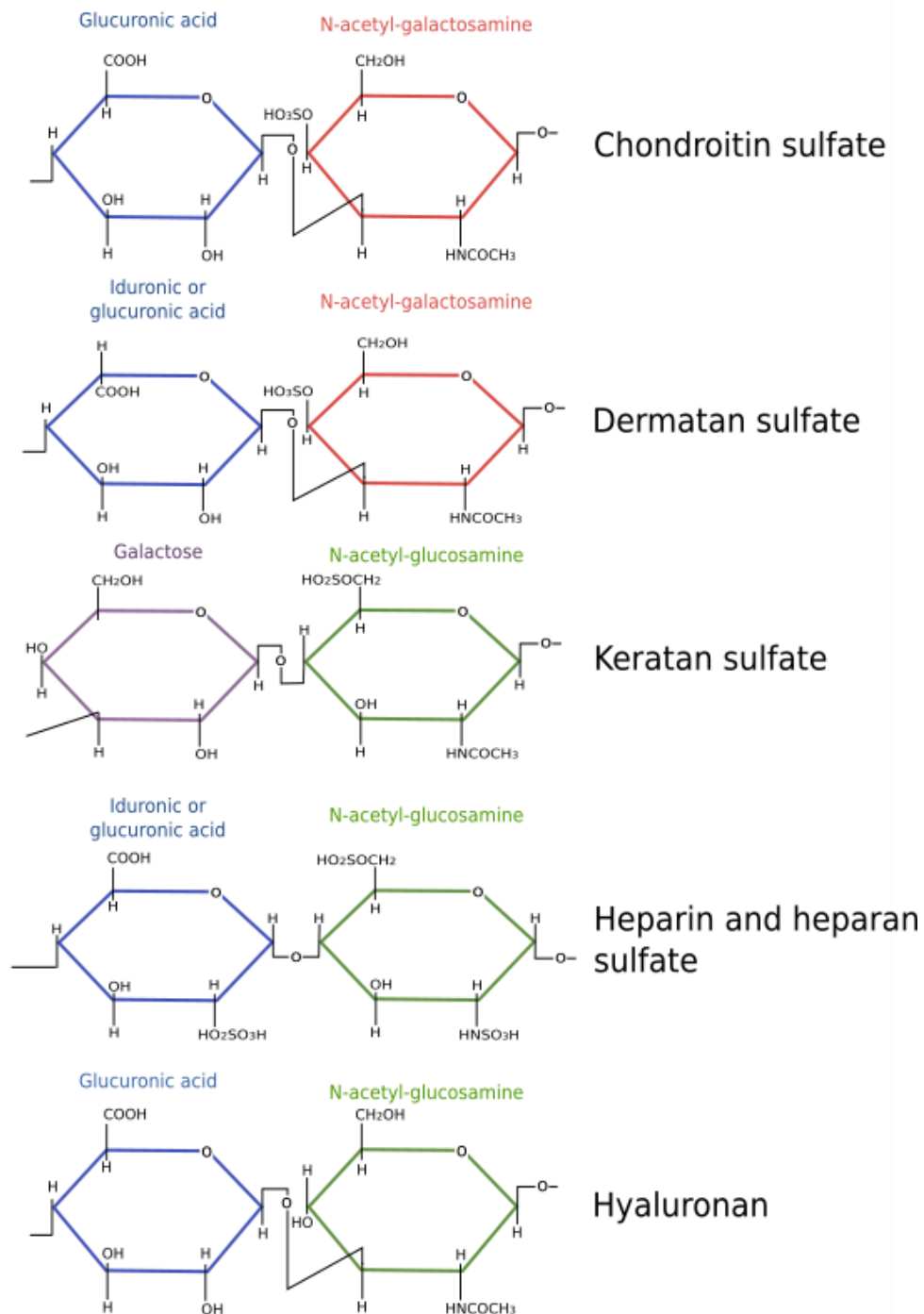


G. Extracellular Matrix

The extracellular space in animal tissue is filled with a gel-like material called the extracellular matrix. It holds cells together and provides a porous pathway for diffusion of O_2 and nutrients to cells. It is composed of a meshwork of heteropolysaccharides and fibrous proteins. The heteropolysaccharides are called glycosaminoglycans. They contain many negative charges which cause the molecule to be extended and give the solution a high viscosity. The glycosaminoglycan hyaluronic acid or hyaluronate has alternating units of D-glucuronic acid and N-acetylglucosamine joined by β 1,3 and β 1,4 bonds with molecular weights over a million.

Molecules of this type are lubricants in joints and give strength to the extracellular matrix of cartilage and tendons. Other glycosaminoglycans

usually contain a uronic acid, NAG, or N-acetylgalactosamine, and some contain sulfate groups.



Proteoglycans are long glycosaminoglycans bound non-covalently to numerous protein molecules, which in turn are covalently bound to smaller

glycosaminoglycan molecules such as chondroitin sulfate. The covalent bonds between carbohydrate and protein are mainly through serine residues. There can be as many as 150 polysaccharide chains per protein molecule, with about 100 protein molecules per one molecule of extended hyaluronate. This interacts with fibrous proteins like collagen and elastin, forming a cross-linked network to which cells attach (via proteins) and along which cell migration is directed.

Representative homopolysaccharides				
homopolysaccharide	sugar component	linkage	function	sources
cellulose	glucose	β , 1 $\rightarrow$ 4	structural	throughout plant kingdom
amylose	glucose	α , 1 $\rightarrow$ 4	food storage	starches, especially corn, potatoes, rice
chitin	N-acetylglucosamine	β , 1 $\rightarrow$ 4	structural	insect and crustacean skeleton
inulin	fructose	β , 2 $\rightarrow$ 1	food storage	artichokes, chicory
xylan	xylose	β , 1 $\rightarrow$ 4	structural	all land plants
glycogen	glucose	α , 1 $\rightarrow$ 4, 6 $\leftarrow$ 1, α	food storage	liver and muscle cells of all animals
amylopectin	glucose	α , 1 $\rightarrow$ 4, 6 $\leftarrow$ 1, α	food storage	starches, especially corn, potatoes, rice
dextran	glucose	α , 1 $\rightarrow$ 6, 4 $\leftarrow$ 1, α	unknown	primarily bacterial
agar*	galactose	α , 1 $\rightarrow$ 3	structural	seaweeds

Representative heteropolysaccharides			
heteropolysaccharide	component sugars	functions	distribution
hyaluronic acid	D-glucuronic acid and <i>N</i> -acetyl-D-glucosamine	lubricant, shock absorber, water binding	connective tissue, skin
chondroitin-4-sulfate*	D-glucuronic acid and <i>N</i> -acetyl-D-galactosamine-4-O-sulfate	calcium accumulation, cartilage and bone formation	cartilage
heparin*	D-glucuronic acid, <i>L</i> -iduronic acid, <i>N</i> -sulfo-D-glucosamine	anticoagulant	mast cells, blood
gamma globulin*	<i>N</i> -acetyl-hexosamine, D-mannose, D-galactose	antibody	blood
blood group substance*	D-glucosamine, D-galactosamine, <i>L</i> -fucose, D-galactose	blood group specificity	cell surfaces, especially red blood cells

TOPIC THREE

NUCLEIC ACIDS

Introduction

Nucleotides are the monomer units of nucleic acids. Just as amino acids are the building blocks of proteins, so nucleotides are the building blocks of nucleic acids. In addition to being nucleic acid components, nucleotides themselves function in energy storage and transfer during metabolic reactions. They also function as cofactors for certain enzymes and as chemical signals in the response to hormones.

Biomedical Importance

Knowledge of the structure and function of nucleic acids is essential in Understanding genetics and many aspects of pathophysiology as well as the genetic basis of disease.

The discovery that genetic information is coded along the length of a polymeric molecule composed of only four types of monomeric units was one of the major scientific achievements of the 20th century. This polymeric molecule, **deoxyribonucleic acid (DNA)**, is the chemical basis of heredity and is organized into genes, the fundamental units of genetic information. The basic information pathway—that is, DNA, which directs the synthesis of RNA, which in turn both directs and regulates protein synthesis—has been elucidated. Genes do not function autonomously; rather their replication and function are controlled by various gene products, often in collaboration with components of various signal transduction pathways. Knowledge of the structure and function of nucleic acids is essential in understanding genetics and many aspects of pathophysiology as well as the genetic basis of disease.

Genetic diseases occur when errors are introduced into the genes that DNA carries; those errors create faulty RNA, which creates faulty proteins that don't function the way they're supposed to. Cancer is caused by damage to

DNA or interference with the mechanisms for its replication or repair. By understanding nucleic acids and their mechanics of action, we can understand how diseases occur and, eventually, how to cure them

Structure of Nucleotides

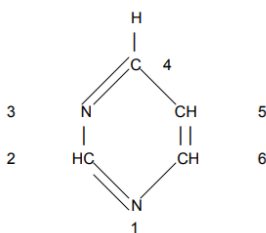
Nucleotides have three component parts, a nitrogenous base, a sugar and a phosphate group.

I. Nitrogenous Bases

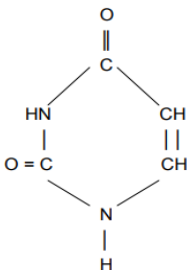
These are heterocyclic compounds containing carbon and nitrogen in a 5 or 6-membered ring. The rings are unsaturated and have aromatic character. They are weak bases that are uncharged at pH = 7. There are two major types, pyrimidines and purines.

A. Pyrimidines

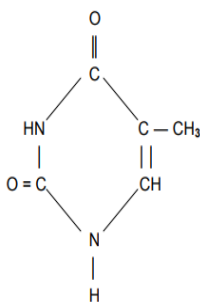
Pyrimidines are based on the parent compound pyrimidine, which contains one heterocyclic ring.



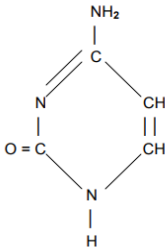
Numerous derivatives can be made, but three pyrimidines are predominant in nucleotides and nucleic acids.



uracil - U
2,4-dioxypyrimidine



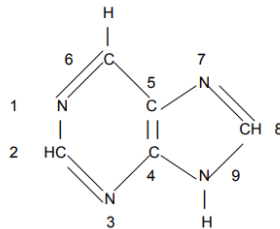
thymine - T
5-methyl-2,4-dioxypyrimidine



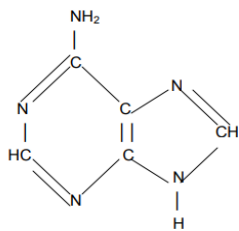
cytosine - C
4-amino -2-oxypyrimidine

B. Purines

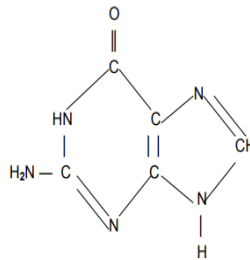
Purines are based on the parent compound purine, which is a derivative of pyrimidine. Purines consist of a pyrimidine ring fused with an imidazole ring to give two heterocyclic rings, one 5-membered and one 6-membered.



Two important purines are found in nucleic acids.

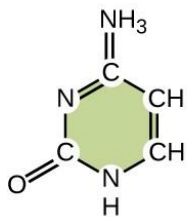


adenine - A
6-aminopurine

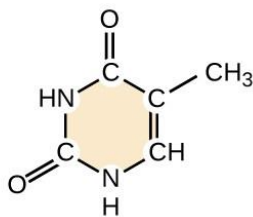


guanine - G
2-amino -6-oxopurine

pyrimidines

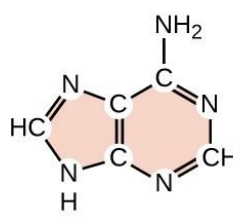


cytosine
C

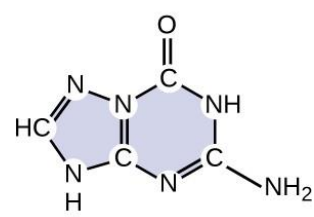


thymine (in DNA)
T

purines



adenine
A

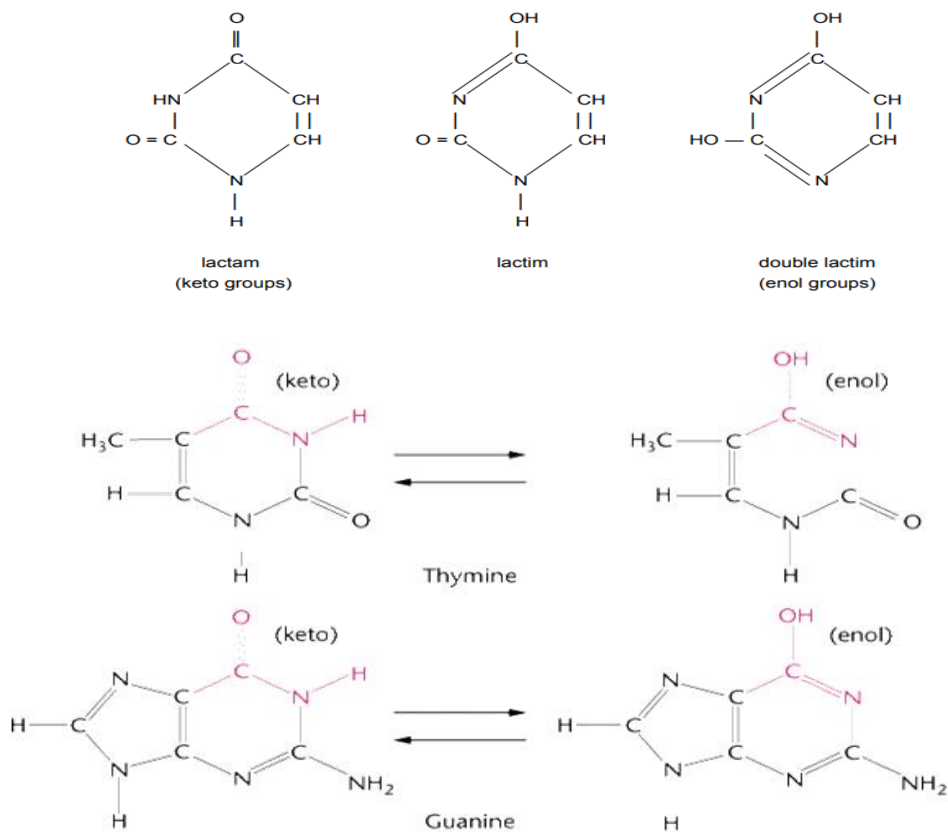


guanine
G

General Features

- a. Both purines and pyrimidines have aromatic character and so are stable despite the presence of numerous double bonds.
- b. Both types are weak bases but will be uncharged at pH = 7.0.
- c. Pyrimidines are planar in shape. Purines are nearly planar with a slight pucker.
- d. Both types have a great capacity for forming hydrogen bonds with the NH, NH₂, N, C = O groups.
- e. The forms shown above are the predominant forms at pH = 7.0. However, alternate tautomeric forms can exist, with the amounts of each form varying with pH.

The tautomers of uracil are shown below.

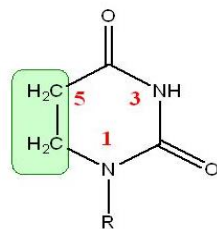


Tautomeric Shifts I: keto and enol bases

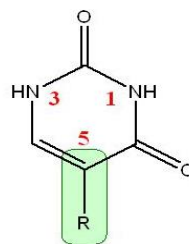
Standard **keto** and rare **enol** forms of **T** & **G** differ by a **spontaneous proton shift** an **H** nucleus between the adjacent **C** and **N** molecules. The molecular formula does not change: the keto and enol forms are therefore described as **tautomers** of each other.

Rare bases

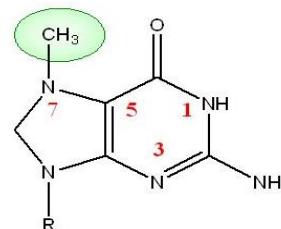
In addition to the five major bases, other unusual bases do exist and are sometimes found in nucleic acids. Often these rare bases have extra methyl groups. They still share the general characteristics of the more common bases.



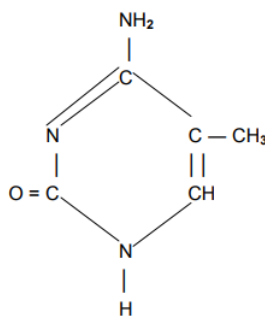
Dihydrouridine
(DHU)



pseudouridine
(ψ)



7-Methylguanosine
(^mG)



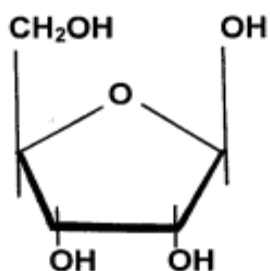
5-methylcytosine



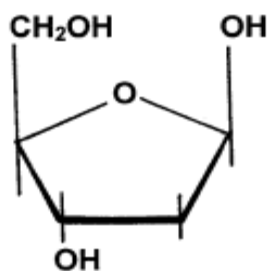
The second component of a nucleotide is a **sugar**. When a sugar is joined to a nitrogenous base, the resulting structure is called a **nucleoside**.

II. Sugars

Two types of sugars are found in nucleotides, both based on D-ribose.



β -D-ribofuranose

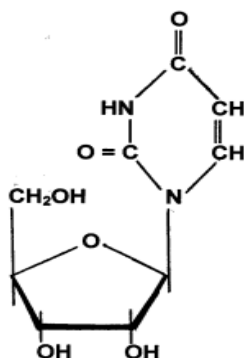


2-deoxy - β -D-ribofuranose
(lacks OH at 2 position)

A. Nucleoside

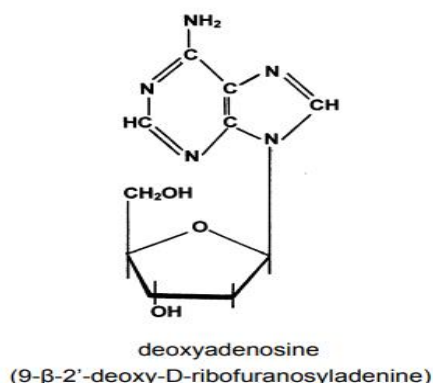
Structure

When the sugar is linked to a nitrogenous base, a nucleoside is produced. The bond forms between the C-1 of the sugar and one of the nitrogens of the rings of the nitrogenous base. This is an N-glycosidic bond. It is relatively stable to alkali but is susceptible to acid hydrolysis (purines more than pyrimidines). The orientation of the bond is β . In pyrimidines the bond will be 1,1. In purines the bond is 1,9. To distinguish between the numbering of the sugar and the numbering of the nitrogenous base, primes are added to the numbers that refer to the sugar.



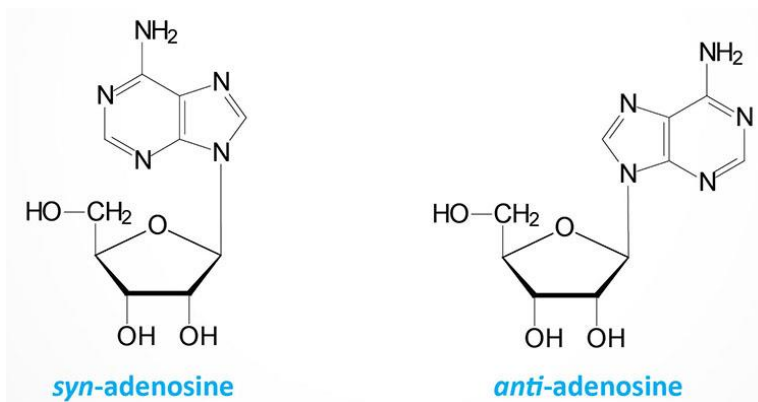
uridine
(1- β -D-ribofuranosyluracil)

The N-glycosidic bond links the 1 position of the pyrimidine to the 1' position of the ribose.



The N-glycosidic bond links the 9 position of the purine to the 1' position of the deoxyribose.

The sugar and the base are both nearly planar and lie at right angles to each other. Two conformations, syn and anti, are possible due to rotation around the N-glycosidic bond. **Due to steric hindrance, the anti conformation is favoured.**



B. Nomenclature

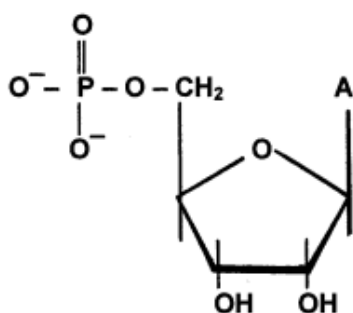
To simplify the names, nucleosides are named by adding -dine or -sine to the name of the base. Deoxy as a prefix signifies the deoxyribose sugar. No prefix indicates the ribose sugar.

	Base	Nucleoside
Ribose in RNA	Adenine	Adenosine
	Guanine	Guanosine
	Cytosine	Cytidine
	Uracil	Uridine
Deoxyribose in DNA	Adenine	Deoxyadenosine
	Guanine	Deoxyguanosine
	Cytosine	Deoxycytidine
	Thymine	Deoxythymidine

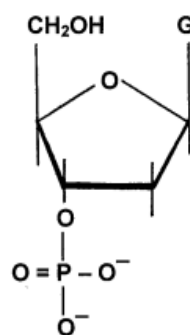
III. Nucleotides

The third component of nucleotides is **one or more phosphate** groups. When a phosphate group is added to a nucleoside, the resulting compound is a nucleotide.

Nucleotides are distinguished by two features, where the phosphate group is attached and how many phosphate groups are attached. The phosphate group will be attached to the ribose ring, but can vary as to which OH group it is joined to. The 5'-position is by far the most common, with the 3'-position also occurring.



adenosine -5'-monophosphate

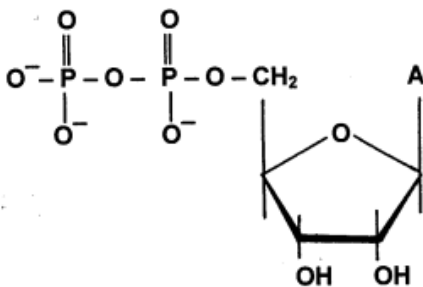


deoxyguanosine-3'-monophosphate

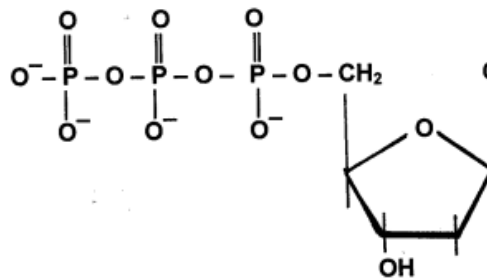
'Mono' is used to indicate the presence of one phosphate group. If no number is included in the name, such as adenosine monophosphate, then the assumption is that the phosphate group is at the 5' position. At pH = 7.0, phosphate will be ionized and carry a negative charge of -2.

Number of phosphates

Many nucleotides have two or three phosphate groups, usually on the same carbon, although mixed 3',5' phosphates can occur. Adenosine-5'-diphosphate charge = -3



adenosine-5'-diphosphate
charge = -3



deoxycytidine-5'-triphosphate
charge = -4

Nomenclature

To simplify the names, each compound is given an abbreviation.

- The first capital letter indicates the base,
- The second letter indicates the number of phosphates, and
- The third letter indicates phosphate.
- A small 'd' in front of the abbreviation signifies a deoxyribonucleotide. Without the 'd', it is assumed the compound is a ribonucleotide. 5' ribonucleotides (NMP = nucleoside monophosphate)

5' ribonucleotides (NMP = nucleoside monophosphate)

<u>Base</u>	<u>NMP</u>	<u>NDP</u>	<u>NTP</u>
A	AMP	ADP	ATP
G	GMP	GDP	GTP
C	CMP	CDP	CTP
U	UMP	UDP	UTP

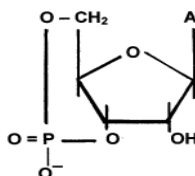
5'-deoxyribonucleotides (dNMP = deoxyribonucleoside monophosphate)

<u>Base</u>	<u>dNMP</u>	<u>dNDP</u>	<u>dNTP</u>
A	dAMP	dADP	dATP
G	dGMP	dGDP	dGTP
C	dCMP	dCDP	dCTP
T	dTMP	dTDP	dTTP

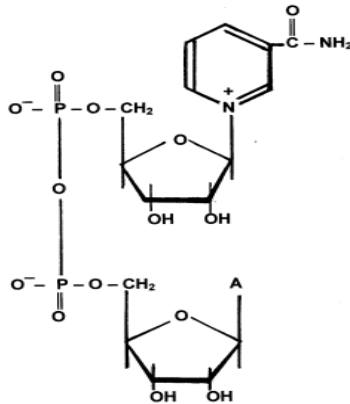
Because of the negative charges, nucleotides are often found in the cell complexed with divalent ions like Mg^{2+} . In the cell, phosphate groups are easily transferred by enzymes, so there is a constant interconversion among mono, di, and triphosphates. Free nucleosides and bases are found in cells only in very low levels.

Function of Nucleotides

- ATP and other nucleotides are used for energy storage and transfer in cells as they lose or gain phosphate groups.
- They are also involved in enzyme regulation, participating in both allostery and covalent modification.
- Unusual nucleotides, such as 3',5'-cyclic AMP (cAMP) mediate the action of hormones.



- Nucleic acids: Nucleosides and heterocyclic chemistry, including Some nucleotides function as enzyme cofactors, like nicotinamide adenine dinucleotide (NAD⁺).



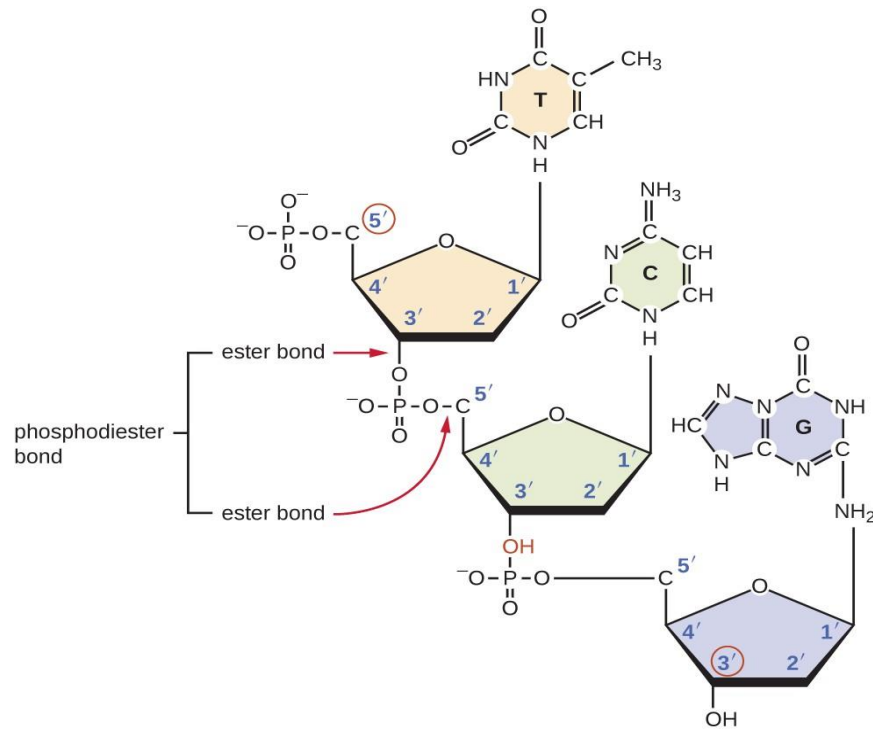
NUCLEIC ACIDS

Nucleic acids are long polymers of nucleotides. They function in the storage, transmission and usage of genetic information. There are two basic types of nucleic acids, DNA (deoxyribonucleic acid) and RNA (ribonucleic acid). DNA is the genetic material of the cell. It contains the cell's genes, which in turn determine all the characteristics of a cell. RNA (there are several types) functions in using the genetic information to make proteins which determine the nature of the cell.

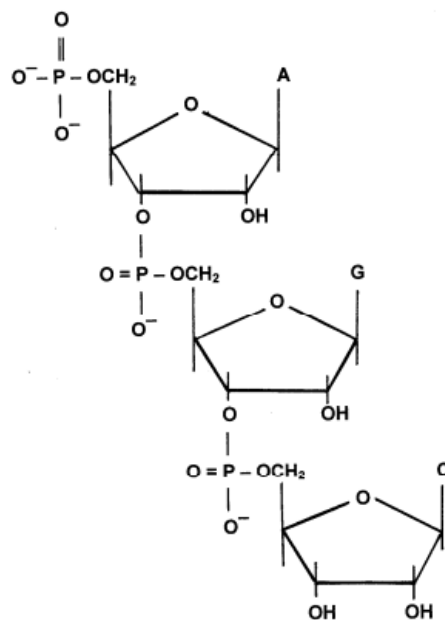
Structure of Nucleic Acids

A. Phosphodiester Bonds

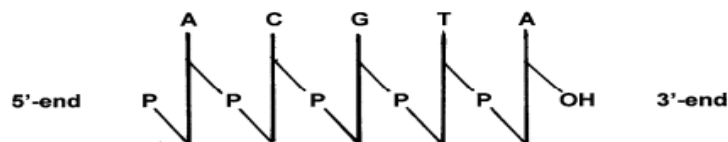
Nucleotides are joined together by phosphodiester bonds into a nucleic acid. The sequence in which the different nucleotides are joined is the primary structure of a nucleic acid.



Phosphodiester bonds form between the phosphate group attached to the 5' carbon of one nucleotide and the hydroxyl group of the 3' carbon in the next nucleotide, bringing about polymerization of nucleotides into nucleic acid strands. Note the 5' and 3' ends of this nucleic acid strand.



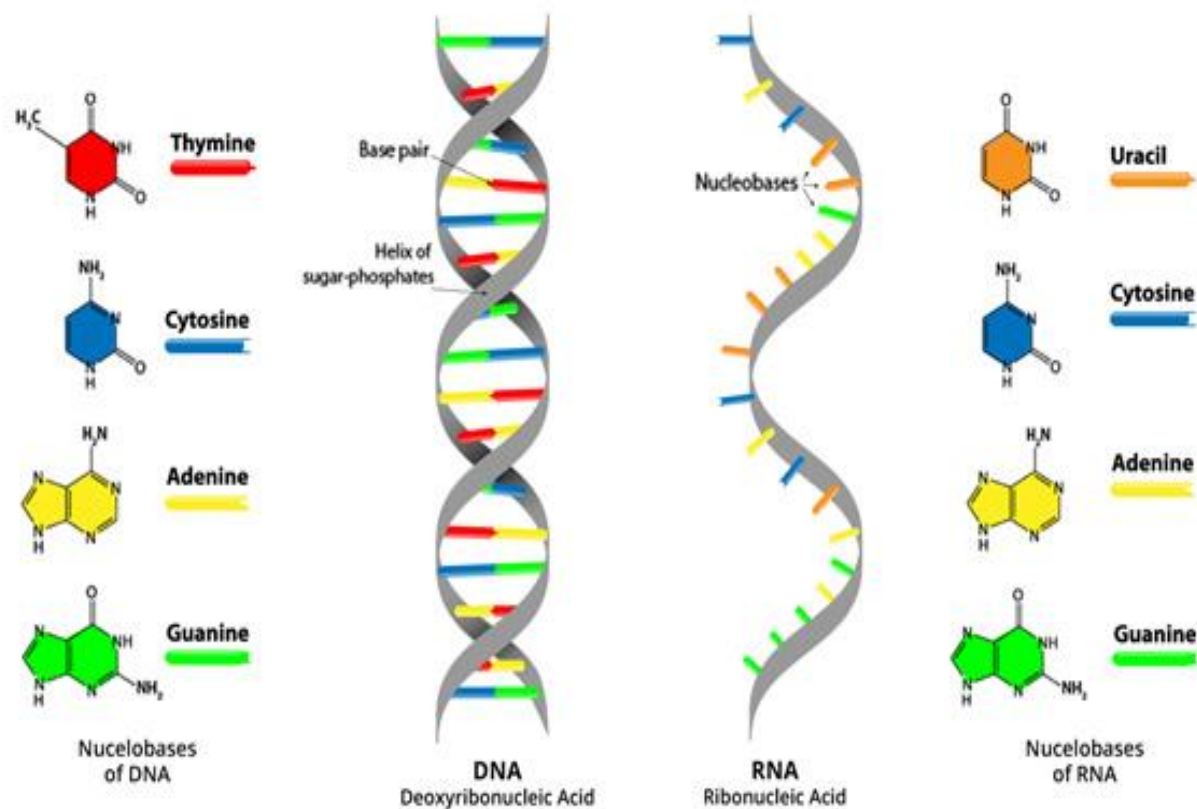
- The bond is always a 3',5'-phosphodiester bond, linking the 5' carbon of one sugar ring to the 3' carbon of the next.
- RNA contains 2'-OH groups while DNA would lack them.
- The covalent backbone thus consists of alternating phosphate and sugar groups.
- The bases are linked to the sugars but are not part of the phosphodiester bonds.
- The phosphate groups will be negatively charged at pH = 7.0.
- The two ends of the chain are not identical. One end has a 5'-phosphate and the other has a 3'-OH. Thus the nucleic acid chain is said to have directionality or polarity.
- The backbone is very hydrophilic while the bases are more hydrophobic.
- Nucleic acids range in size from less than 100 nucleotides to more than 10,000,000 nucleotides.
- To consolidate the structure of nucleic acids, several notations can be used.



or pApCpGpTpA or pACGTA

- The 5'-end by convention is on the left.

Some nucleic acids, mainly double-stranded DNA, form more complex structures. The major form of secondary structure is the double-helix. The double-helix is a spiral shape, somewhat similar to the α -helix, except that two chains are involved.



Each turn of the helix is 34 Å or 3.4 nm containing 10 base-pairs. (Each base-pair occupies 0.34 nm.) The width is 2.0 nm (20 Å). The helix is right-handed. The two chains have opposite directionality and so are antiparallel. The hydrophilic phosphates and sugars are on the outside where they can H-bond to water and where repulsion between the two negatively-charged chains can be minimized. The more hydrophobic bases are protected in the interior. The bases lie perpendicular to the helix and parallel to each other. The helix contains two grooves on the exterior of unequal size, called the major groove and the minor groove which also spiral around the structure. The two chains are held together by non-covalent forces. The first force is the relative hydrophilic vs. hydrophobic nature of the different parts of the nucleic acid. The second force is base stacking. The bases within a strand are parallel to each other and there are van der Waals forces, dipole-dipole interactions and hydrophobic interactions between them that stabilize the helix (and have

effects on their electronic interactions and light absorbance). The third force is hydrogen bonding. Only certain base combinations can hydrogen bond.

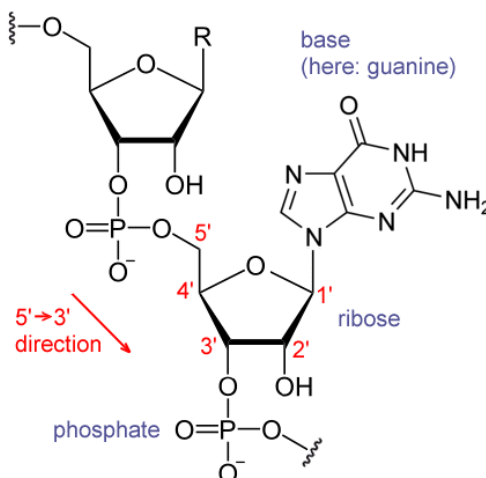
Ribonucleic Acid (RNA)

RNA is a nucleotide polymer consisting of four bases — adenine (A), guanine (G), cytosine (C) and uracil (U)—sugar and phosphate. The information contained by DNA is transferred to the whole body at specific locations through RNA. They are involved in several essential metabolic functions required for living organisms.

Structure of RNA

RNA is a single-strand helical structure. The phosphate group present in the structure makes it a charged molecule (polyanion). A distinguishing feature in the RNA structure is the presence of hydroxyl group at the 2' position of the ribose sugar, which gives it an A-form geometrical structure.

RNA structures have the capability of self-folding, that is, interchain complementary base pairing between the bases of the strand. It leads to the formation of bulges and helices in the RNA strands.



Types and Functions of RNA

RNA molecules can be found in multiple copies and several forms inside the cells. A few major classes of RNA include mRNA, rRNA, tRNA, snRNA, snoRNA, miRNA, XIST, siRNA, tmRNA, and telomerase RNA. Out of all these RNAs, the most well-known and well-studied RNA molecules are: mRNA, rRNA and tRNA. They play diverse roles in organisms including transfer of genetic information during protein synthesis, gene expression, enzymatic activity and storage of genetic information in RNA viruses and viroids

1. mRNA

mRNA refers to messenger RNA. It copies and carries the information, encoded in one or more genes, from the DNA to ribosome for protein formation. Some differences between the RNA molecules of eukaryotes and prokaryotes are mentioned below:

	Eukaryotic mRNA	Prokaryotic mRNA
1.	Mostly monocistronic mRNA (encodes only a single protein) with an average size of 1500-2000 nucleotides.	Polycistronic mRNA that encodes several proteins.
2.	It has a cap-like structure at the 5'-end.	It doesn't have any cap-like structure at the 5'-end.
3.	It has a long poly-A tail at the 3' end that helps to stabilize RNA.	It has a short poly-A tail at the 3'-end that acts as a targeting signal for RNA destruction.
4.	It doesn't have intercistronic regions.	It has intercistronic regions.

5.	It requires extensive processing and transport.	It doesn't require any processing.
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2. rRNA

rRNA molecules are ribosomal RNA. These are the most abundant forms of RNA in a cell, forming 80% of the RNA molecules in eukaryotes. It associates with ribosomes to form a complex structure that moves in a 5' to 3' direction to catalyze the formation of proteins. These molecules play an active role in recognizing the conserved positions of mRNA and tRNA.

In eukaryotes, the four different rRNA molecules include 18s, 5.8s, 28s, and 5s rRNA, whereas, in prokaryotes, it includes 16s, 5s, and 23s rRNA. ^[13]

3. tRNA

tRNA is referred to as transfer RNA or adaptor RNA. It carries individual amino acids into the ribosomes to assemble the growing polypeptide chains. tRNA has clover leaf-like structures containing 70-80 nucleotides. The structure has well-defined stems and loops that make up the acceptor arm, D-stem and loop, anticodon stem and loop, and the T-stem and loop.

Each amino acid has its specific type of tRNA. Each tRNA binds to the specific amino acids and carries them to the growing polypeptide chain. The prokaryotes have 30-45 different tRNA and the eukaryotes contain 50 or more tRNA

DNA and RNA

While DNA and RNA are similar, they have very distinct differences.

Features of DNA and RNA		
	DNA	RNA
Function	Carries genetic information	Involved in protein synthesis

Features of DNA and RNA		
	DNA	RNA
Location	Remains in the nucleus	Leaves the nucleus
Structure	DNA is double-stranded "ladder": sugar-phosphate backbone, with base rungs.	Usually single-stranded
Sugar	Deoxyribose	Ribose
Pyrimidines	Cytosine, thymine	Cytosine, uracil
Purines	Adenine, guanine	Adenine, guanine

One other difference bears mention. There is only one type of DNA. DNA is the heritable information that is passed along to each generation

PROTEINS

Introduction

Proteins are the 'doers' of the cell. They are huge in number and variety and diverse in structure and function, serving as both the structural building blocks and the functional machinery of the cell. Just about every process in every cell requires specific proteins. Let us begin by listing some of the basic cellular processes and the role that proteins play.

Chemical catalysis: Enzymes, which are responsible for catalyzing biological reactions, are the largest functional group of proteins. Whilst there are thousands of different enzymes, all catalyzing different reactions, they do have some features in common and can often be identified as members of a particular family of enzymes.

Mechanical support: Typically, support is provided by proteins, e.g. the cytoskeletal proteins inside the cell and the extracellular matrix proteins outside the cell.

Communication: The signals within and between cells (e.g. cytokines) and the apparatus for recognizing and interpreting or reacting to signals (receptors and transducers) are mainly proteins.

Adhesion: Cell surface proteins mediate contact between cells and between a cell and the extracellular matrix (which is also made up of proteins).

Movement: Proteins generate movement in a cell (motor proteins).

Defence: Antibodies (immunoglobulins) are proteins that recognize specific targets (usually proteins themselves). This facility is critical for an immune response.

Transport: Proteins are key molecules in the transport of substances both within a cell and to and from the cell.

Storage: A number of proteins serve to store small molecules or ions; for example, ferritin binds iron and stores it in the liver.

You will come across many examples in the course of your degree proteins performing the functions outlined above and the molecular basis for various cellular processes will be examined in some detail. The basic principles of protein structure and function, which are reviewed in this course, are crucial to understanding how proteins perform their various roles.

The huge diversity in the functions of proteins is reflected in the specialization of these molecules. As you will see in this course, **every protein optimally performs a particular job and the key to how it does so is its structure.**

The refinement of protein structure and optimization of protein function are driven by evolutionary pressures. Mutations at the DNA level that result in a change in protein structure and function will persist if they enhance survival or are not detrimental to the organism. Proteins come in as many different shapes and sizes as they have functions as will be discussed.

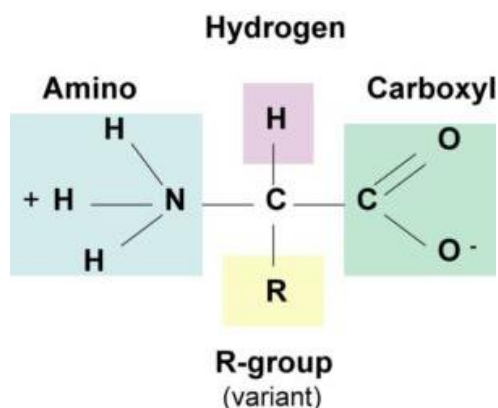
Structure and Stereochemistry of the Acids

Amino acids constitute a group of neutral products clearly distinguished from other natural compounds chemically, mainly because of their ampholytic properties and biochemically mainly because of their role as protein constituents.

An amino acid is a carboxylic acid-containing an aliphatic primary amino group in the α position to the carboxyl group and with a characteristic stereochemistry. Proteins are biosynthesized from 20 amino acids in a system involving strict genetic control. Thus, amino acids are the basic unit of proteins.

More than 300 amino acids are found in nature but only 20 amino acids are standard and present in protein because they are coded by genes. Other amino acids are modified amino acids and called non-protein amino acids. Some are residues modified after a protein has been synthesized by post-

translational modifications; others are amino acids present in living organisms but not as constituents of proteins.



Exceptions are:

- Glycine, which does not have a side chain. Its α -carbon contains two hydrogens.
- Proline, in which the nitrogen is part of a ring.

Thus, each amino acid has an amine group at one end and an acid group at the other and a distinctive side chain. The backbone is the same for all amino acids while the side chain differs from one amino acid to the next.

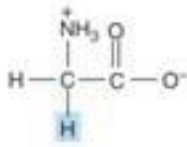
All of the 20 amino acids except glycine are of the L-configuration, as for all but one amino acid the α -carbon is an asymmetric carbon. Because glycine does not contain an asymmetric carbon atom, it is not optically active and thus, it is neither D nor L.

Classification of amino acids on the basis of R-group

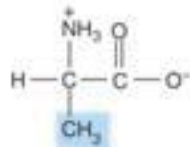
1. **Nonpolar, aliphatic amino acids:** The R groups in this class of amino acids are nonpolar and hydrophobic. Glycine, Alanine, Valine, leucine, Isoleucine, Methionine, Proline.

Nonpolar, aromatic amino acids: Phenylalanine, tyrosine, and tryptophan, with their aromatic side chains, are relatively nonpolar (hydrophobic). All can participate in hydrophobic interactions.

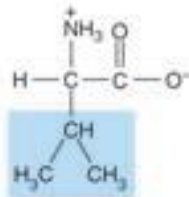
Nonpolar Amino Acids



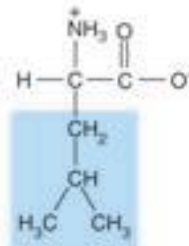
Glycine



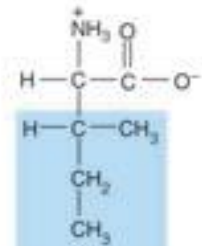
Alanine



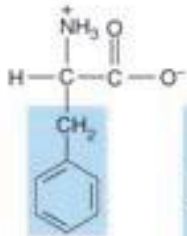
Valine



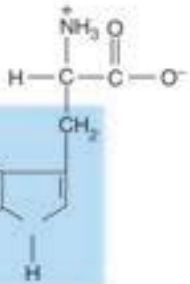
Leucine



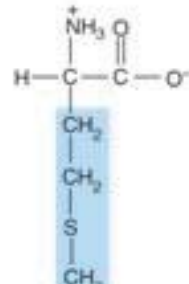
Isoleucine



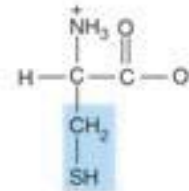
Phenylalanine



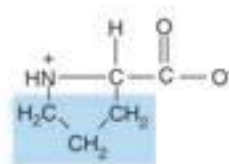
Tryptophan



Methionine

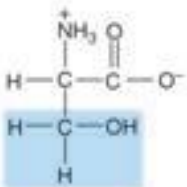


Cysteine

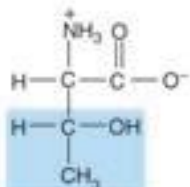


Proline

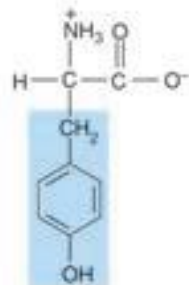
Polar Amino Acids



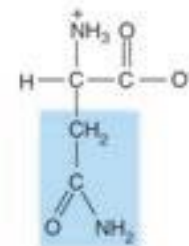
Serine



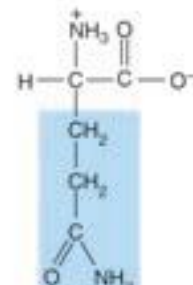
Threonine



Tyrosine

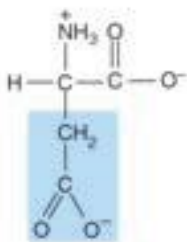


Asparagine

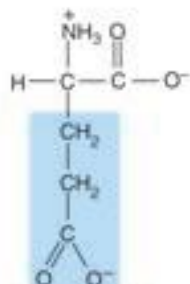


Glutamine

Acidic Amino Acids

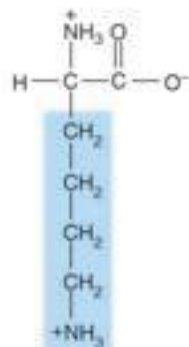


Aspartate

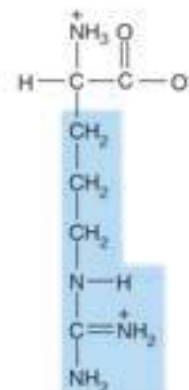


Glutamate

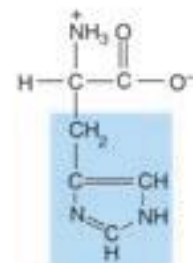
Basic Amino Acids



Lysine



Arginine



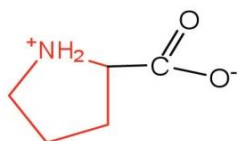
Histidine

2. **Polar,- Uncharged amino acids & charged:** The R groups of these amino acids are more soluble in water, or more hydrophilic, than those of the nonpolar amino acids, because they contain functional groups that form hydrogen bonds with water. This class of amino acids includes serine, threonine, cysteine, asparagine, and glutamine.
3. **Acidic amino acids:** Amino acids in which R-group is acidic or negatively charged. Glutamic acid and Aspartic acid
4. **Basic amino acids:** Amino acids in which R-group is basic or positively charged. Lysine, Arginine, Histidine

Note:

Proline classified as an imino acid, its α -amine is a secondary amine with its a nitrogen having two covalent bonds to carbon (to the α -carbon and side chain carbon), rather than primary amine Proline is the only amino acid which is a secondary amine, in that the nitrogen atom is attached to the alpha-carbon by a chain of three other carbon atoms

Incorporation of amino nitrogen into a five membered ring constrains rotational freedom around $-N\alpha -C\alpha$ -bond in proline to specific rotational angle, reduces structural flexibility of polypeptide regions containing proline

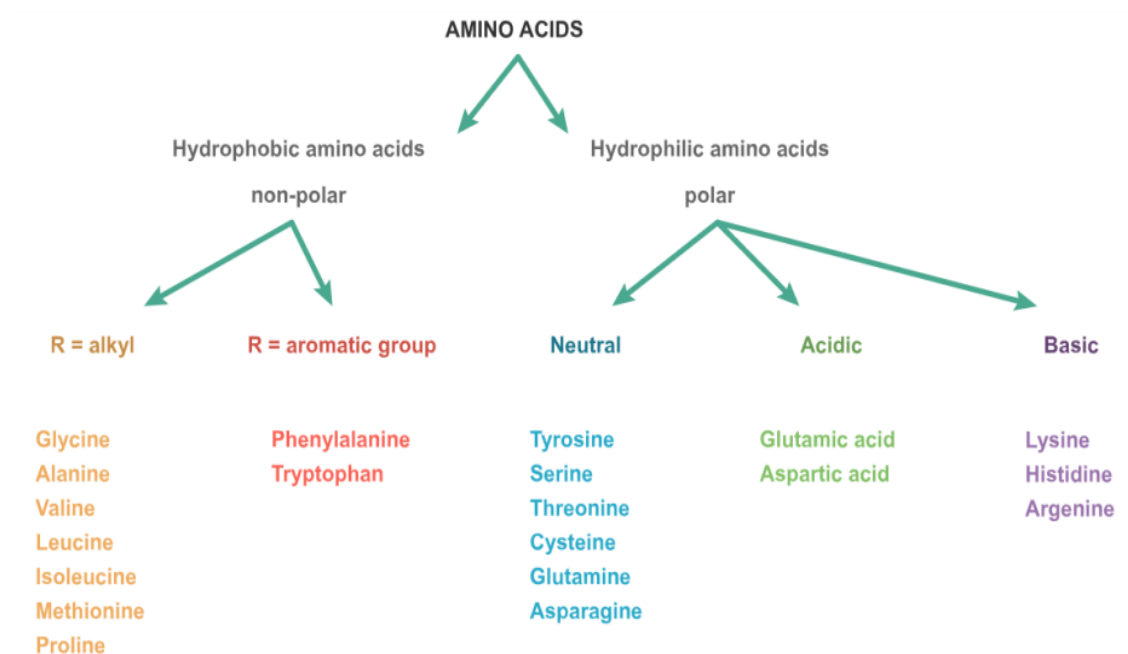


The distinctive cyclic structure of proline's side chain gives proline an exceptional conformational rigidity compared to other amino acids. It also affects the rate of peptide bond formation between proline and other amino acids. When proline is bound as an amide in a peptide bond, its nitrogen is not bound to any hydrogen, meaning it cannot act as a hydrogen bond donor, but can be a hydrogen bond acceptor

The exceptional conformational rigidity of proline affects the secondary structure of proteins near a proline residue and may account for proline's higher prevalence in the proteins of thermophilic organisms. Proline is

commonly found as the **first** residue of an **alpha helix** and also in the **edge** strands of **beta sheets**. Proline is also commonly found in **turns** (another kind of secondary structure) and aids in the formation of beta turns.

In Summary:



Classification of amino acids on the basis of nutrition

Humans can synthesize about half (11) of the amino acids needed to make proteins. They include: Arginine, glutamine, tyrosine, cysteine, glycine, proline, serine, ornithine, alanine, asparagine, and aspartate. Six of the nonessential amino acids (arginine, cysteine, glutamine, glycine, proline, and tyrosine) are considered **conditionally essential** as dietary supplementation may be required during the course of an illness or in children. Other amino acids, called the essential amino acids, must be provided in the diet. The ten essential amino acids are the following: arginine (Arg) valine (Val) methionine (Met) leucine (Leu) threonine (Thr) phenylalanine (Phe) histidine (His) isoleucine (Ile) lysine (Lys) tryptophan (Trp).

Proteins that provide all the essential amino acids in about the right proportions for human nutrition are called **complete proteins**. Examples of

complete proteins are those in meat, fish, milk, and eggs. About 50 g of complete protein per day is adequate for adult humans. Proteins that are severely deficient in one or more of the essential amino acids are called **incomplete proteins**. If the protein in a person's diet comes mostly from one incomplete source, the amount of human protein that can be synthesized is limited by the amounts of the deficient amino acids. Plant proteins are generally incomplete. Rice, corn, and wheat are all deficient in lysine. Rice also lacks threonine and corn also lacks tryptophan. Beans, peas and other legumes have the most complete proteins among the common plants, but they are deficient in methionine.

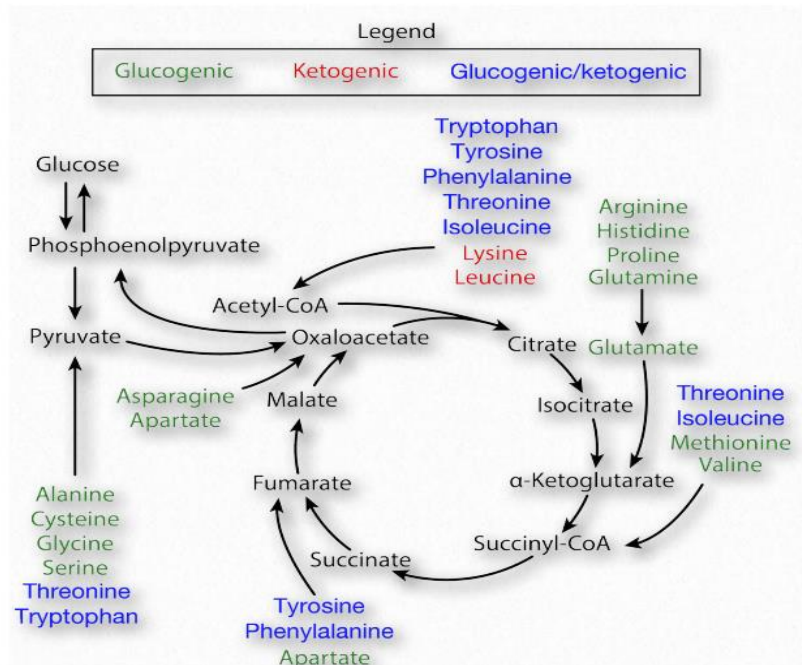
Vegetarians can achieve an adequate intake of the essential amino acids if they eat many different plant foods. Plant proteins can be chosen to be complementary, with some foods supplying amino acids that others lack. An alternative is to supplement the vegetarian diet with a rich source of complete protein such as milk or eggs.

Essential	Conditionally Non-Essential	Non-Essential
Histidine	Arginine	Alanine
Isoleucine	Cystine	Asparagine
Leucine	Glutamine	Aspartate
Lysine	Glycine	Glutamate
Methionine	Proline	Serine
Phenylalanine	Tyrosine	
Threonine		
Tryptophan		
Valine		

Classification of amino acids on the basis of the metabolic fate

Glucogenic amino acids	Glucogenic and ketogenic	Ketogenic amino acids
Alanine, Arginine, Asparagine, Aspartate Asparagine, Cysteine, Methionine Glutamate, Glutamine, Glycine, Histidine Proline, Serine, Threonine,Valine	Tyrosine Isoleucine Phenylalanine Tryptophan	Leucine Lysine

1. **Glucogenic amino acids:** These amino acids serve as precursors gluconeogenesis for glucose formation. Glycine, alanine, serine, aspartic acid, asparagine, glutamic acid, glutamine, proline, valine, methionine, cysteine, histidine, and arginine.
2. **Ketogenic amino acids:** These amino acids breakdown to form ketone bodies. Leucine and Lysine.
3. **Both glucogenic and ketogenic amino acids:** These amino acids breakdown to form precursors for both ketone bodies and glucose. Isoleucine, Phenylalanine, Tryptophan and tyrosine.



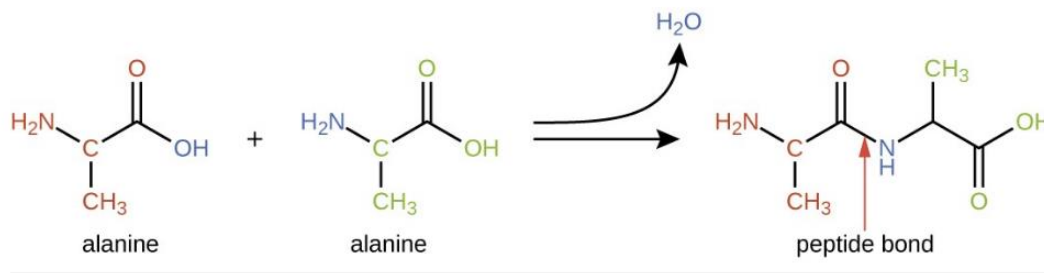
PROTEINS

Proteins are one of the most abundant organic molecules in living systems and have the most diverse range of functions of all macromolecules. The sequence and the number of amino acids ultimately determine the protein's shape, size, and function. Each amino acid is attached to another amino acid by a covalent bond, known as a peptide bond, which is formed by a dehydration reaction. The carboxyl group of one amino acid and the amino group of the incoming amino acid combine, releasing a molecule of water. The resulting bond is the peptide bond.

- A peptide is a short-chain made up of **amino acid** which, together with other peptides, forms a protein.
- The number of amino acids in a peptide can range from two amino acids to fifty amino acids.
- Based on the number of amino acids present in the peptide, peptides are of many types; peptides with ten or fewer amino acids are termed oligopeptides, and the peptides with more than ten amino acids are termed polypeptides.

- Polypeptides with around 100 amino acids are then considered proteins.

Each polypeptide has a free amino group at one end. This end is called the **N terminal** or the amino terminal and the other end has a free carboxyl group, also known as the C or **carboxyl terminal**. While the terms polypeptide and protein are sometimes used interchangeably, a polypeptide is technically a polymer of amino acids, whereas the term protein is used for a polypeptide or polypeptides that have combined together, often have bound non-peptide prosthetic groups, **have a distinct shape and have a unique function**. After protein synthesis (translation), most proteins are modified. These are known as post-translational modifications. They may undergo cleavage, phosphorylation or may require the addition of other chemical groups. Only after these modifications is the protein completely functional.



PROTEIN STRUCTURE

As discussed earlier, the shape of a protein is critical to its function. For example, an enzyme can bind to a specific substrate at a site known as the active site. If this active site is altered because of local changes or changes in overall protein structure, the enzyme may be unable to bind to the substrate. To understand how the protein gets its final shape or conformation, we need to understand the four levels of protein structure: primary, secondary, tertiary, and quaternary.

Primary structure includes all the covalent bonds in a protein, both peptide bonds and disulfide bonds. However, rotation is possible around many of these

covalent bonds, **so there are a large number of possible 3-dimensional shapes that a protein can assume.** The spatial arrangement of atoms in a protein is called its conformation. Despite the almost limitless number of possible conformations of a protein, each protein will have a specific, unique 3-dimensional structure. The higher levels of protein structure describe this conformation with increasing complexity. **There are several important points about conformation.**

A. Function of a protein depends upon its conformation. Protein molecules that lose their proper shape will not be able to function properly.

B. The proper conformation for a protein is often (but not always) the one which is the most thermodynamically stable. What shape is most stable depends on what amino acids are in the protein? Thus conformation depends upon the amino acid sequence. It is a goal of protein research to be able to deduce protein conformation from a protein's primary structure. Unfortunately, given the large number of possible conformations and the difficulty of estimating energies and stabilities, this cannot be done at present. Limited predictions about the structure of regions in proteins can currently be made.

C. Conformation is maintained and held together mainly by non-covalent forces. (Disulfide bonds and other covalent crosslinks also help to maintain protein shape.) The various non-covalent forces include the following types.

- Hydrogen bonds can form in many ways between certain R groups of amino acids, between portions of the peptide backbone, and between polar amino acids and the surrounding water molecules. Many polar amino acids tend to cluster on the outside of a protein molecule where they can interact with the solvent.
- Ionic interactions involve attractions between opposite charges. Two oppositely charged amino acids can form an ionic bond known as a salt

bridge. Other electrostatic interactions involve induced or permanent dipoles.

- Van der Waals forces are weak attractions between close, uncharged atoms. A random variation in the electron cloud of one atom creates a momentary dipole that induces an opposite dipole in another atom and causes an attraction.
- Hydrophobic interactions occur when two or more hydrophobic groups cluster together and so avoid interaction with water. There is no actual attraction between the non-polar groups, but rather the stability comes from the thermodynamic favorability of keeping these groups from water where they cause the water to assume a highly structured solvation layer. Non-polar amino acids tend to cluster in the interior of the protein where water is excluded. While non-covalent forces are weak, there are a large number of them in a given protein. The amino acid sequence will dictate how the chain must be spatially arranged in order to maximize these forces.

D. Types of conformations

Secondary structure refers to the arrangement of neighboring amino acids, which often occurs in a regular, repeating structure.

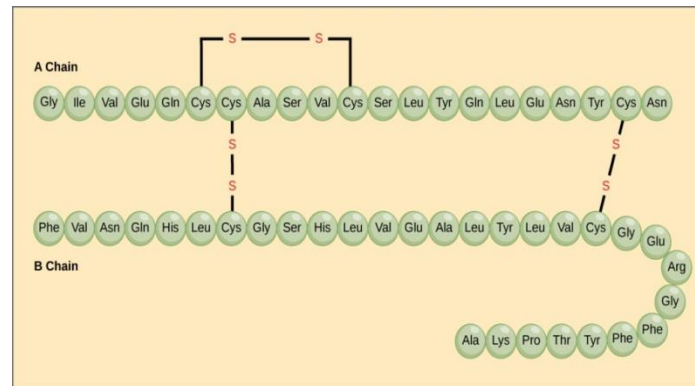
Tertiary structure refers to the complete 3-dimensional structure of the polypeptide chain.

Quaternary structure occurs in multimeric proteins and refers to the spatial arrangement of the subunits.

PRIMARY STRUCTURE

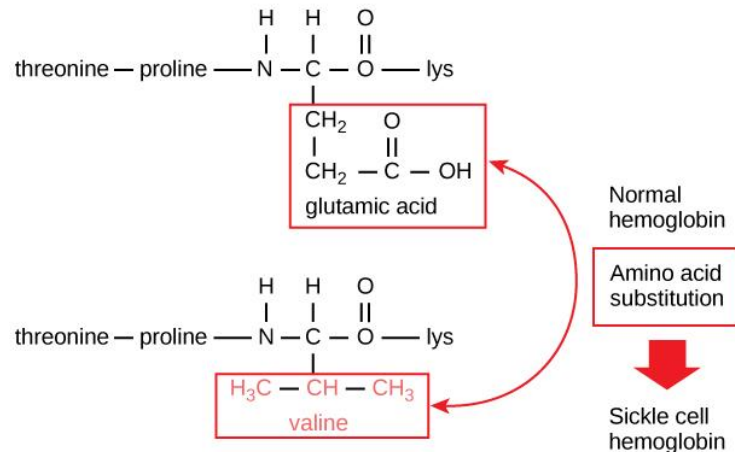
The unique sequence of amino acids in a polypeptide chain is its primary structure. For example, the pancreatic hormone insulin has two polypeptide chains, A and B, and they are linked together by disulfide bonds. The N terminal amino acid of the A chain is glycine, whereas the C terminal amino

acid is asparagine (Figure below). The sequences of amino acids in the A and B chains are unique to insulin.



The unique sequence for every protein is ultimately determined by the gene encoding the protein. A change in nucleotide sequence of the gene's coding region may lead to a different amino acid being added to the growing polypeptide chain, causing a change in protein structure and function. In sickle cell anemia, the hemoglobin β chain (a small portion of which is shown in the Figure) has a single amino acid substitution, causing a change in protein structure and function.

Specifically, the amino acid glutamic acid is substituted by valine in the β chain. What is most remarkable to consider is that a hemoglobin molecule is made up of **two alpha chains** and **two beta chains** that each consist of about 150 amino acids. The molecule, therefore, has about 600 amino acids. The structural difference between a normal hemoglobin molecule and a sickle cell molecule which dramatically decreases life expectancy is a single amino acid of the 600. What is even more remarkable is that those 600 amino acids are encoded by three nucleotides each and the mutation is caused by a single base change (point mutation), 1 in 1800 bases.

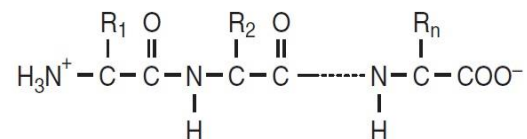


The beta chain of hemoglobin is 147 residues in length, yet a single amino acid substitution leads to sickle cell anemia. In normal hemoglobin, the amino acid at position seven is glutamate. In sickle cell hemoglobin, this glutamate is replaced by a valine.

Because of this change of one amino acid in the chain, hemoglobin molecules form long fibres that distort the biconcave, or disc-shaped, red blood cells and assume a crescent or “sickle” shape, which clogs arteries. This can lead to myriad serious health problems such as breathlessness, dizziness, headaches and abdominal pain for those affected by this disease.

Primary structure

(always written with the free amino group to the left):

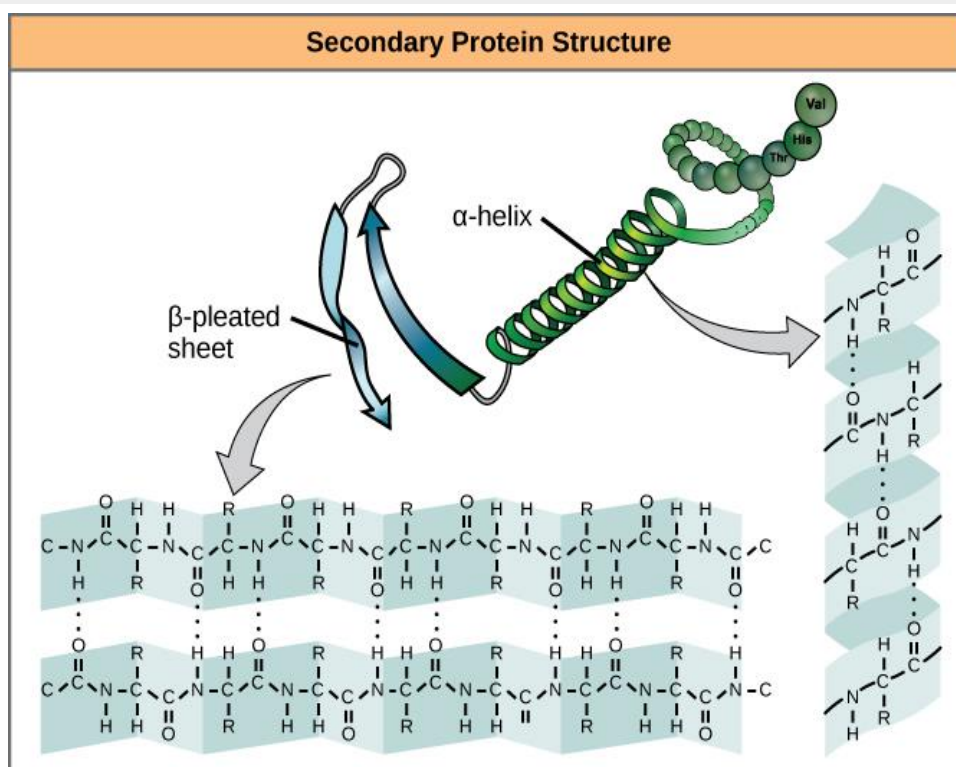


Secondary Structure

The sequence of amino acids will cause the protein to have areas of its shape conforming to one of 3 shapes, alpha-helical, beta-pleated sheet or woven (sometimes called turns). These are **secondary structures**. A molecule has a characteristic size and shape. The function of many molecules depends upon their shape

The local folding of the polypeptide in some regions gives rise to the secondary structure of the protein. The most common are the α -helix and β -pleated sheet structures as in the Figure. Both structures are the α -helix structure—the helix held in shape by hydrogen bonds. The hydrogen bonds form between the oxygen atom in the carbonyl group in one amino acid and another amino acid that is four amino acids farther along the chain.

The α -helix and β -pleated sheet are secondary structures of proteins that form because of hydrogen bonding between carbonyl and amino groups in the peptide backbone. Certain amino acids have a propensity to form an α -helix, while others have a propensity to form a β -pleated sheet.



Every helical turn in an alpha helix has 3.6 amino acid residues. The R groups (the variant groups) of the polypeptide protrude out from the α -helix chain. In the β -pleated sheet, the “pleats” are formed by hydrogen bonding between atoms on the backbone of the polypeptide chain. The R groups are attached to the carbons and extend above and below the folds of the pleat. The pleated

segments align parallel or antiparallel to each other, and hydrogen bonds form between the partially positive nitrogen atom in the amino group and the partially negative oxygen atom in the carbonyl group of the peptide backbone. The α -helix and β -pleated sheet structures are found in most globular and fibrous proteins and they play an important structural role.

Proteins can be divided into two categories with regard to 2° structure.

- 1. Globular proteins** have a very compact 3-dimensional structure, within which exist different areas of 2° structure. Different globular proteins have very different amounts of 2° structure types.

	<u>α</u>	<u>β</u>	<u>irregular</u>
myoglobin	78%	0%	22%
chymotrypsin	14%	45%	41%
lysozyme	40%	12%	48%

- 2. Fibrous proteins** have long, extended structures consisting often of just one main type of 2° structure. There are several categories of fibrous proteins.

- a. α -keratins**

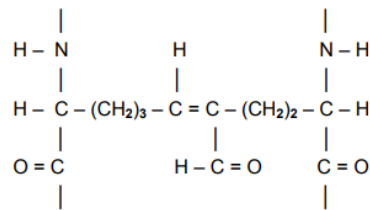
α -keratins make up hair, wool, feathers, nails, skin, horns, etc. The structure of α -keratins is a right-handed α -helix. They contain many hydrophobic residues, making them insoluble and water-repellent. Adjacent chains of α -helix are linked by disulfide bonds. The more disulfide bonds, the stronger the structure is. In hair, two α -helices interact to form coils, two coils combine to make a protofilament and four protofilaments make one filament. These are packaged and wound to form macrofibrils, many of which make up a hair strand. Disulfide cross-links occur at several levels, producing a strong structure. In the case of hair, the structure is flexible but does not stretch.

- b. β -keratins**

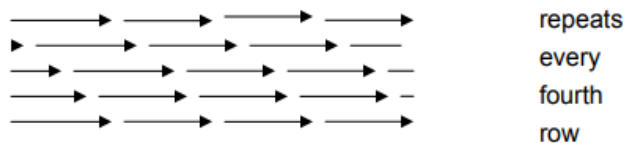
β -keratins include silk and spider webs that are flexible but do not stretch. The structure is an antiparallel β -sheet with a high percentage of small amino acids (Gly, Ala). There are no disulfide bonds. The small R groups allow pairs of the β -sheets to stack together, forming an ordered array.

c. Collagen

Collagen (existing in several different types) is found in tendons, cartilage, etc. Collagen has a unique left-handed helix with 3.3 amino acids/ turn. The helix is more extended than an α -helix. Collagen contains 35% Gly and 21% proline plus hydroxyproline, usually in a repeating sequence of Gly - X - Pro (Hyp). Three such chains (~ 1000 amino acids) are wound together in a right-handed triple helix held together by hydrogen bonds and covalent crosslinks between oxidized lysines.



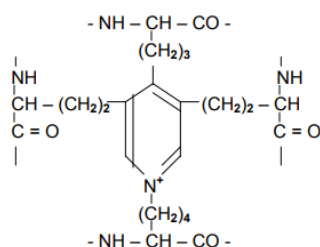
A collagen fibril consists of many triple helices (tropocollagen molecules) arranged head to tail in a staggered array. Covalent cross-links produce a fibril that does not stretch.



d. Elastin

Elastin is also found in connective tissue and it will stretch. The basic unit is tropoelastin which is 800 amino acids, probably arranged in some kind of coil or helix. It is rich in Gly, Ala, and Lys, but low in Pro. Lysines form covalent

cross-links where four lysines join to form desmosine, which stretches reversibly.

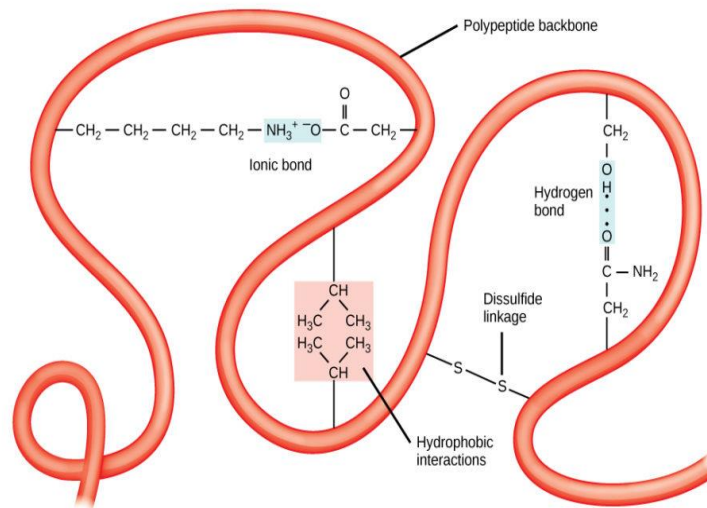


Tertiary Structure

The unique three-dimensional structure of a polypeptide is its tertiary structure. This structure is in part due to chemical interactions at work on the polypeptide chain. Primarily, the interactions among R groups create the complex three-dimensional tertiary structure of a protein. The nature of the R groups found in the amino acids involved can counteract the formation of the hydrogen bonds described for standard secondary structures. For example, R groups with like charges are repelled by each other and those with unlike charges are attracted to each other (ionic bonds).

When protein folding takes place, the hydrophobic R groups of nonpolar amino acids lay in the interior of the protein, whereas the hydrophilic R groups lay on the outside. The former types of interactions are also known as hydrophobic interactions. Interaction between cysteine side chains forms disulfide linkages in the presence of oxygen, the only covalent bond forming during protein folding.

The tertiary structure of proteins is determined by a variety of chemical interactions. These include hydrophobic interactions, ionic bonding, hydrogen bonding and disulfide linkages. All of these interactions, weak and strong, determine the final three-dimensional shape of the protein. When a protein loses its three-dimensional shape, it may no longer be functional.



Quaternary Structure

In nature, some proteins are formed from several polypeptides, also known as subunits, and the interaction of these subunits forms the quaternary structure. Weak interactions between the subunits help to stabilize the overall structure.

For example, insulin (a globular protein) has a combination of hydrogen bonds and disulfide bonds that cause it to be mostly clumped into a ball shape. Insulin starts out as a single polypeptide and loses some internal sequences in the presence of post-translational modification after the formation of the disulfide linkages that hold the remaining chains together. Silk (a fibrous protein), however, has a β -pleated sheet structure that is the result of hydrogen bonding between different chains.

LIPIDS

Introduction

The lipids are a heterogeneous group of compounds, including fats, oils, steroids, waxes, and related compounds, which are related more by their physical than by their chemical properties. Lipids are a class of compounds distinguished by their insolubility in water and solubility in nonpolar solvents. Lipids are important in biological systems because they form the cell membrane, a mechanical barrier that divides a cell from the external environment. Lipids also **provide energy for life** and several essential vitamins are lipids. Lipids can be divided in two major classes, non-saponifiable lipids and saponifiable lipids.

A **non-saponifiable** lipid cannot be broken up into smaller molecules by hydrolysis, which includes triglycerides, waxes, phospholipids, and sphingolipids. A **saponifiable** lipid contains one or more ester groups allowing it to undergo hydrolysis in the presence of an acid, base, or enzyme. Non-saponifiable lipids include **steroids, prostaglandins, and terpenes**. Within these two major classes of lipids, there are several specific types of lipids important to life, including fatty acids, triglycerides, glycerophospholipids, sphingolipids, and steroids. Each of these categories can be further broken down. **Nonpolar lipids**, such as triglycerides, are used for energy storage and fuel. **Polar lipids**, which can form a barrier with an external water environment, are used in membranes. Polar lipids include glycerophospholipids and sphingolipids. Fatty acids are important components of all of these lipids

Biological Roles of Lipids

Lipids have the common property of being relatively insoluble in water and soluble in nonpolar solvents such as ether and chloroform. They are important dietary constituents not only because of their high energy value but also because of the fat-soluble vitamins and the essential fatty acids contained in the fat of natural foods. Fat is stored in adipose tissue, where it also serves

as a thermal insulator in the subcutaneous tissues and around certain organs. Nonpolar lipids act as electrical insulators, allowing rapid propagation of depolarization waves along myelinated nerves. Combinations of lipid and protein (lipoproteins) are important cellular constituents, occurring both in the cell membrane and in the mitochondria, and serving also as the means of transporting lipids in the blood. Knowledge of lipid biochemistry is necessary in understanding many important biomedical areas, e.g., obesity, diabetes mellitus, atherosclerosis, and the role of various polyunsaturated fatty acids in nutrition and health.

CLASSIFICATION OF LIPIDS

Lipids are classified as follows:

a) Based on structure and composition

1. Simple lipids: Esters of fatty acids with various alcohols.

(a) Fats: Esters of fatty acids with glycerol. Oils are fats in the liquid state. (b) Waxes: Esters of fatty acids with higher molecular weight monohydric alcohols.

2. Complex lipids: Esters of fatty acids containing groups in addition to an alcohol and a fatty acid.

- i. **Phospholipids:** Lipids containing, in addition to fatty acids and an alcohol, a phosphoric acid residue. They frequently have nitrogen containing bases and other substituents, e.g, in glycerophospholipids the alcohol is glycerol and in sphingophospholipids the alcohol is sphingosine.
 - ii. **Glycolipids** (glycosphingolipids): Lipids containing a fatty acid, sphingosine, and carbohydrate.
 - iii. **Other complex lipids:** Lipids such as sulfolipids and aminolipids. Lipoproteins may also be placed in this category.
- #### **3. Precursor and derived lipids:** These include fatty acids, glycerol, steroids, other alcohols, fatty aldehydes, and ketone bodies, hydrocarbons, lipid

soluble vitamins, and hormones. Because they are uncharged, acylglycerols (glycerides), cholesterol, and cholesteryl esters are termed neutral lipids

b) Based on their function

- a) Energy-storage lipids—A fat, triacylglycerols or triglycerides
- b) Membrane lipids-phospholipids, sphingoglycolipids, and cholesterol
- c) Emulsification lipids-bile acids, soaps and detergents
- d) Chemical messenger lipids-steroid hormones, eicosanoids, and prostaglandins
- e) Protective-coating lipids-biological waxes
- f) Fat-soluble vitamins

Functions of Lipids in the Body:

Storing Energy

The excess energy from the food we eat is digested and incorporated into adipose tissue or fatty tissue. Most of the energy required by the human body is provided by carbohydrates and lipids. "Carbohydrates", glucose is stored in the body as glycogen. While glycogen provides a ready source of energy, lipids primarily function as an energy reserve. As you may recall, glycogen is quite bulky with heavy water content, thus the body cannot store too much for long. Alternatively, fats are packed together tightly without water and store far greater amounts of energy in a reduced space.

A fat gram is densely concentrated with energy—it contains more than double the amount of energy than a gram of carbohydrate. Carbohydrates or proteins only provide 4 kcal/g of energy, fatty acids provide more than twice the energy per unit weight of 9 kcal/ g. This is one reason why a high-fat diet can lead to obesity.

Energy is needed to power the muscles for all the physical work and play an average person or child engages in. For instance, the stored energy in muscles

propels an athlete down the track, spurs a dancer's legs to showcase the latest fancy steps and keeps all the moving parts of the body functioning smoothly. Unlike other body cells that can store fat in limited supplies, fat cells are specialized for fat storage and are able to expand almost indefinitely in size. An overabundance of adipose tissue can result in undue stress on the body and can be detrimental to your health.

Regulating and Signaling

Triacylglycerols control the body's internal climate, maintaining a constant temperature. Those who don't have enough fat in their bodies tend to feel cold sooner, are often fatigued and have pressure sores on their skin from fatty acid deficiency. Triacylglycerols also help the body produce and regulate hormones. For example, adipose tissue secretes the hormone leptin, which regulates appetite. In the reproductive system, fatty acids are required for proper reproductive health; women who lack proper amounts may stop menstruating and become infertile. Omega-3 and omega-6 essential fatty acids help regulate cholesterol and blood clotting and control inflammation in the joints, tissues and bloodstream. Fats also play important functional roles in sustaining nerve impulse transmission, memory storage, and tissue structure. More specifically in the brain, lipids are focal to brain activity in structure and in function. They help form nerve cell membranes, insulate neurons and facilitate the signaling of electrical impulses throughout the brain.

Insulating and Protecting

Did you know that up to 30 percent of body weight is comprised of fat tissue? Some of this is made up of visceral fat or adipose tissue surrounding delicate organs. Vital organs such as the heart, kidneys and liver are protected by visceral fat. The composition of the brain is outstandingly 60 percent fats, demonstrating the major structural role that fat serves within the body. You may be most familiar with subcutaneous fat, or fat underneath the skin. This

blanket layer of tissue insulates the body from extreme temperatures and helps keep the internal climate under control. It pads our hands and buttocks and prevents friction, as these areas frequently come in contact with hard surfaces. It also gives the body the extra padding required when engaging in physically demanding activities such as ice- or roller skating, horseback riding, or snowboarding.

Aiding Digestion and Increasing Bioavailability

The dietary fats in the foods we eat break down in our digestive systems and begin the transport of precious micronutrients. By carrying fat-soluble nutrients through the digestive process, intestinal absorption is improved. This improved absorption is also known as increased bioavailability. Fat-soluble nutrients are especially important for good health and exhibit a variety of functions. Vitamins A, D, E, and K—the fat-soluble vitamins—are mainly found in foods containing fat. Some fat-soluble vitamins (such as vitamin A) are also found in naturally fat-free foods such as green leafy vegetables, carrots, and broccoli. These vitamins are best absorbed when combined with foods containing fat. Fats also increase the bioavailability of compounds known as phytochemicals, which are plant constituents such as lycopene (found in tomatoes) and beta-carotene (found in carrots). Phytochemicals are believed to promote health and well-being. As a result, eating tomatoes with olive oil or salad dressing will facilitate lycopene absorption. Other essential nutrients, such as essential fatty acids, are constituents of the fats themselves and serve as building blocks of a cell.

Note that removing the lipid elements from food also takes away the food's fat-soluble vitamin content. When products such as grain and dairy are processed, these essential nutrients are lost. Manufacturers replace these nutrients through a process called enrichment.

Smell and Taste

Fat contains dissolved compounds that contribute to mouth-watering aromas and flavours and increase palatability of food. Fat also adds texture to food. Baked foods are supple and moist. Frying foods locks in flavour and lessens cooking time. How long does it take you to recall the smell of your favorite food cooking? What would a meal be without that savory aroma to delight your senses and heighten your preparedness for eating a meal?

Fat plays another valuable role in nutrition. Fat contributes to satiety or the sensation of fullness. When fatty foods are swallowed the body responds by enabling the processes controlling digestion to retard the movement of food along the digestive tract, thus promoting an overall sense of fullness. Oftentimes before the feeling of fullness arrives, people overindulge in fat-rich foods, finding the delectable taste irresistible. Indeed, the very things that make fat-rich foods attractive also make them a hindrance to maintaining a healthful diet.

STRUCTURES OF LIPIDS

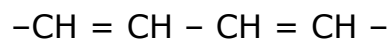
Lipids comprise a broad class of many chemically distinct compounds

I. Fatty Acids

A fatty acid is a molecule characterized by the presence of a carboxyl group attached to a long hydrocarbon chain. Therefore these are molecules with a formula $R-COOH$ where R is a hydrocarbon chain. Fatty acids can be said to be carboxylic acids, and come in two major varieties.

- **Saturated fatty acids do not have any double bonds.** A fatty acid is saturated when every carbon atom in the hydrocarbon chain is bonded to as many hydrogen atoms as possible (the carbon atoms are saturated with hydrogen). Saturated fatty acids are solids at room temperature. Animal fats are a source of saturated fatty acids. In addition, fatty acids pack easily and form rigid structures (e.g., fatty acids are found in membranes).

- Unsaturated fatty acids can **have one or more double bonds** along its hydrocarbon chain. A fatty acid with one double bond is called monounsaturated. If it contains two or more double bonds, we say that the fatty acid is polyunsaturated. The melting point of a fatty acid is influenced by the number of double bonds that the molecule contains and by the length of the hydrocarbon tail. The more double bonds it contains, the lower the melting point. As the length of the tail increases, the melting point increases. Plants are the source of unsaturated fatty acids



Unsaturated fatty acid chain



Saturated fatty acid chain

Naturally occurring fatty acids have cis bonds. Trans-fatty acids are created artificially using a process called hydrogenation. A trans-fatty acid has a trans configuration rather than cis configuration at each double bond. This causes the molecule to straighten. These two stereoisomers can be distinguished in the following way:

- In a cis stereoisomer, two similar groups attached to the carbon double bond are found on the same side.
- In a trans stereoisomer, two similar groups attached to the carbon double bond are found on opposite sides.

CLASSIFICATION OF FATTY ACIDS

NONESSENTIAL FATTY ACIDS If a fatty acid can only be obtained from the diet (for humans) then the fatty acid is an essential fatty acid. Two fatty acids cannot be synthesized in the human body and are therefore essential. These are linoleic and linolenic fatty acids, which are both unsaturated. Nonessential fatty acids can be made by the human body and so do not need to be obtained

from diet alone. These are made from carbohydrates and proteins or from other fatty acids.

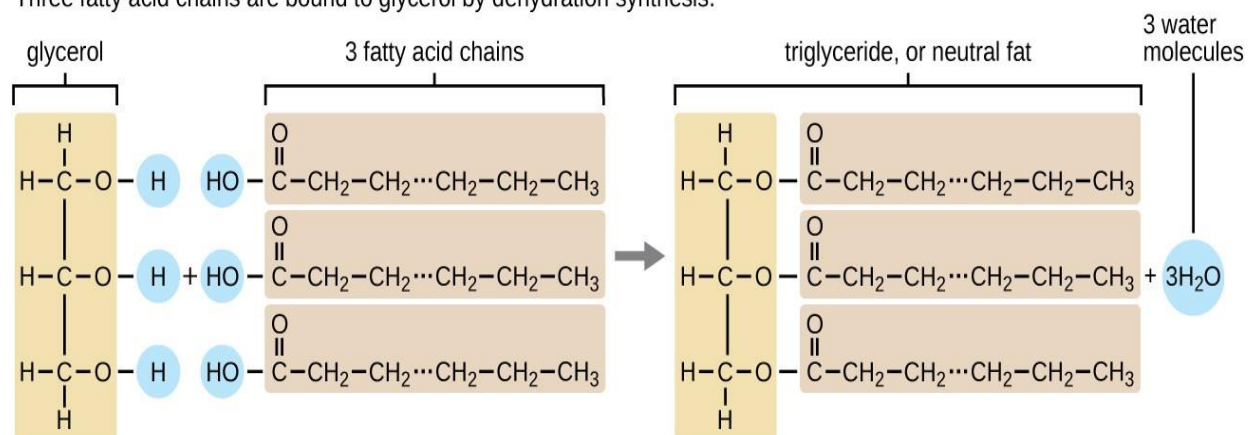
Fatty acids are aliphatic carboxylic acids. Fatty acids occur mainly as esters in natural fats and oils but do occur in the unesterified form as free fatty acids, a transport form found in the plasma. Fatty acids that occur in natural fats are usually straight-chain derivatives containing an even number of carbon atoms. The chain may be saturated (containing no double bonds) or unsaturated (containing one or more double bonds).

Most naturally occurring unsaturated fatty acids have cis double bonds. The carbon chains of saturated fatty acids form a zigzag pattern when extended, as at low temperatures. At higher temperatures, some bonds rotate, causing chain shortening, which explains why biomembranes become thinner with increases in temperature. A type of geometric isomerism occurs in unsaturated fatty acids, depending on the orientation of atoms or groups around the axes of double bonds, which do not allow rotation. If the acyl chains are on the same side of the bond, it is cis-, as in oleic acid; if on opposite sides, it is trans-, as in elaidic acid, the trans isomer of oleic acid. Naturally occurring unsaturated long-chain fatty acids are nearly all of cis configuration. Thus, oleic acid has an L shape, whereas elaidic acid remains "straight." Increase in the number of cis double bonds in a fatty acid leads to a variety of possible spatial configurations of the molecule—e.g., arachidonic acid, with four cis double bonds, has "kinks" or a U shape. This has profound significance on molecular packing in membranes and on the positions occupied by fatty acids in more complex molecules such as phospholipids. Trans double bonds alter these spatial relationships. Trans fatty acids are present in certain foods, arising as a by-product of the saturation of fatty acids during hydrogenation, or "hardening," of natural oils in the manufacture of margarine. An additional small contribution comes from the ingestion of

ruminant fat that contains trans fatty acids arising from the action of microorganisms in the rumen

II. Triacylglycerides

Three fatty acid chains are bound to glycerol by dehydration synthesis.



A triglyceride (often called triacylglycerol) is a fatty acid trimer of glycerol. Triglycerides are important for human health in that they provide most of the lipids in our diet. Glycerol has three hydroxyl groups. Fatty acids can be attached at these three sites forming a triglyceride. One important characteristic of a triglyceride is its state at room temperature. The degree of saturation and the length of their chains attached to the glycerol backbone both determine their state at room temperature.

- Short-chain unsaturated triglycerides are liquid at room temperature.
- Long-chain saturated triglycerides are solid at room temperature.

Animal fats (lard) contain a high amount of saturated triglycerides while plant oils (vegetable oil) contain a high amount of unsaturated triglycerides. While neither is healthy when consumed in excess, vegetable oils are far healthier than lard.

Triglycerides are the main storage forms of fatty acids. The triglycerides are esters of the trihydric alcohol glycerol and fatty acids. Mono- and diacylglycerols wherein one or two fatty acids are esterified with glycerol are

also found in the tissues. These are of particular significance in the synthesis and hydrolysis of triacylglycerols.

The role of triglycerides in health

While fat may seem bad, triglycerides play many important roles in the body. For example, triglycerides can be used for energy storage in animals. This food reserve can be called upon during periods of starvation, with the high-calorie content of the fatty acids adding to the value of storing fat and providing much needed energy. In addition, triglycerides can provide insulation for animals in the form of body fat, which allows them to survive in colder temperatures. These two roles played by fat in the body, which arose over eons of evolution, are now deemed undesirable in modern industrialized society where humans no longer face starvation or have to deal directly with cold weather.

III. Waxes

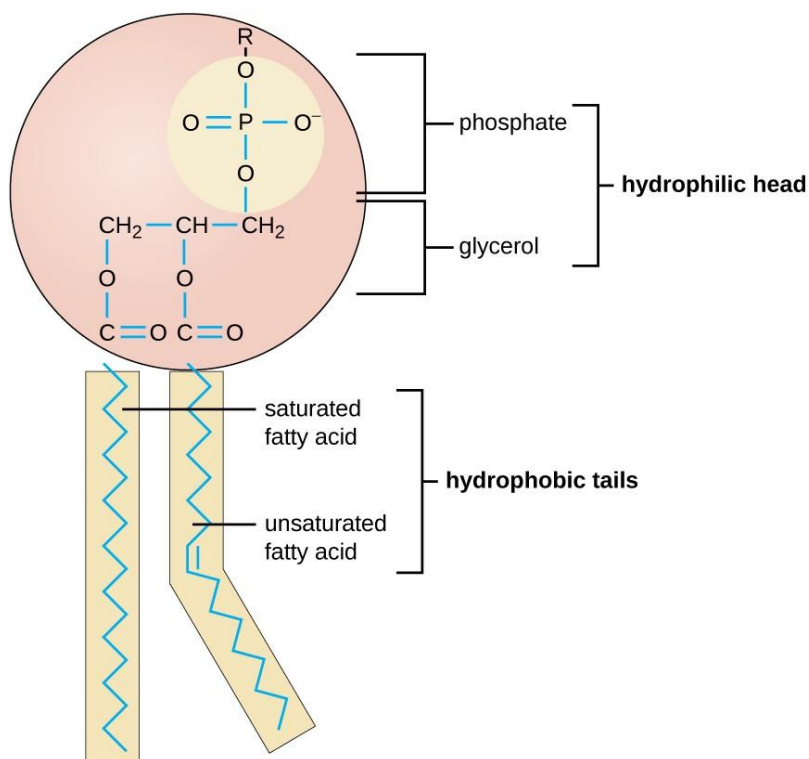
Waxes are esters of long-chain fatty acids (14-36 carbons) with long-chain alcohols (16-30 carbons). Waxes are a type of long chain nonpolar lipid. Natural waxes are typically esters of fatty acids and long chain alcohols. Waxes are synthesized by many animals and plants. Animal wax esters are typically derived from a variety of carboxylic acids and fatty alcohols. The composition of a wax depends not only on the species, but also on the geographic location of the organism. The best known animal wax is beeswax, but other insects secrete waxes as well. A major component of beeswax is the ester myricyl palmitate, which bees use for constructing honeycombs.

Spermaceti is also a wax that occurs in large amounts in the oil of a sperm whale's head. One of its main constituents is cetyl palmitate, an ester of a fatty acid and fatty alcohol. Plant waxes are derived from mixtures of long-chain hydrocarbons containing functional groups such as alkanes, fatty acids, alcohols, diols, ketones, and aldehydes. Plants also use waxes as a protective

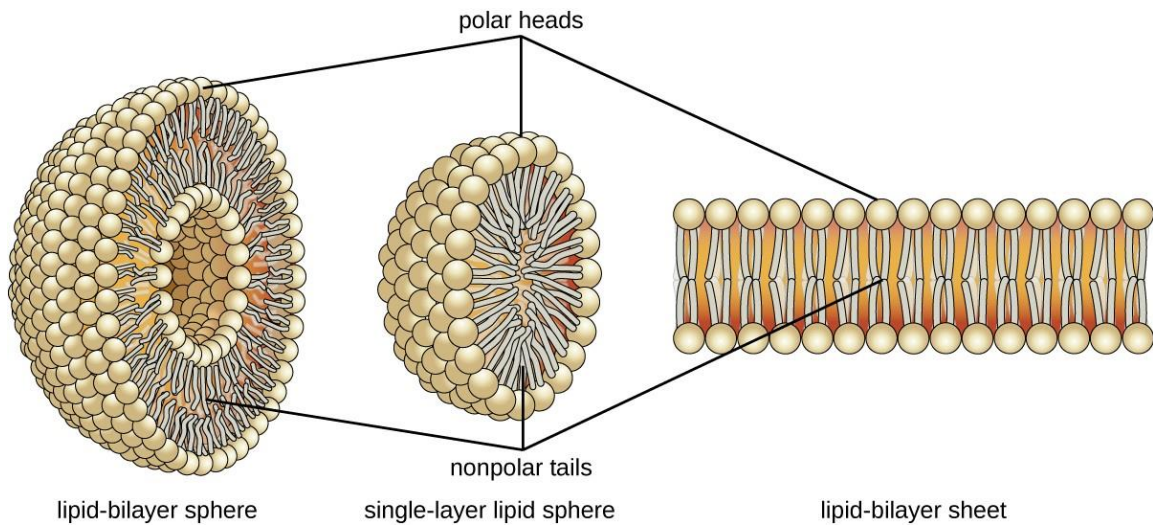
coating to control evaporation and hydration and to prevent them from drying out. Waxes are valuable to both plants and animals because of their hydrophobic nature. This makes them water resistant, which prevents water from sticking on surfaces.

IV. Phospholipids (Glycerophospholipids)

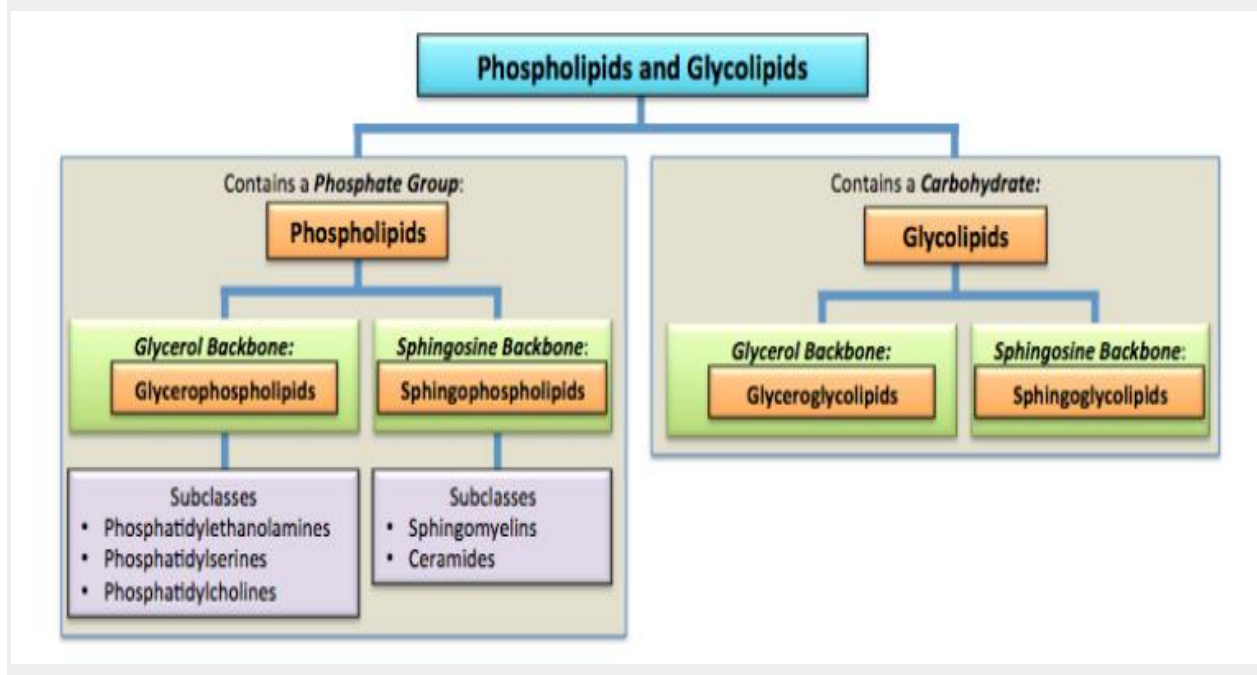
Phospholipids are the main lipid constituents of membranes. Phospholipids may be regarded as derivatives of phosphatidic acid, in which the phosphate is esterified with the OH of a suitable alcohol. Phosphatidic acid is important as an intermediate in the synthesis of triacylglycerols as well as phosphoglycerols but is not found in any great quantity in tissues. Amphipathic lipids self-orient at oil:water interfaces. They form Membranes, Micelles,



This illustration shows a phospholipid with two different fatty acids, one saturated and one unsaturated, bonded to the glycerol molecule. The unsaturated fatty acid has a slight kink in its structure due to the double bond.



Phospholipids tend to arrange themselves in aqueous solution forming liposomes, micelles, or lipid bilayer sheets.



V. Liposomes and Emulsions

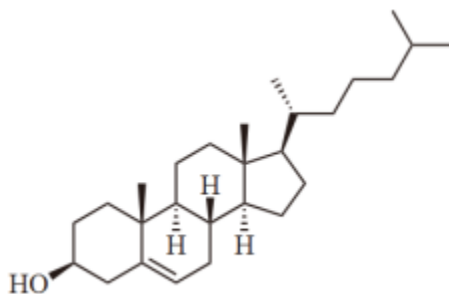
In general, lipids are insoluble in water since they contain a predominance of nonpolar (hydrocarbon) groups. However, fatty acids, phospholipids, sphingolipids, bile salts, and, to a lesser extent, cholesterol contain polar groups. Therefore, part of the molecule is hydrophobic, or water-insoluble; and part is hydrophilic, or water-soluble. Such molecules are described as

amphipathic. They become oriented at oil:water interfaces with the polar group in the water phase and the nonpolar group in the oil phase.

A bilayer of such amphipathic lipids has been regarded as a basic structure in biologic membranes. When a critical concentration of these lipids is present in an aqueous medium, they form micelles. Aggregations of bile salts into micelles and liposomes and the formation of mixed micelles with the products of fat digestion are important in facilitating absorption of lipids from the intestine. Liposomes may be formed by sonicating an amphipathic lipid in an aqueous medium. They consist of spheres of lipid bilayers that enclose part of the aqueous medium. They are of potential clinical use – particularly when combined with tissue specific antibodies – as carriers of drugs in the circulation, targeted to specific organs, eg, in cancer therapy. In addition, they are being used for gene transfer into vascular cells and as carriers for topical and transdermal.

VI. CHOLESTEROL AND ITS IMPORTANCE

Cholesterol is an important lipid found in the cell membrane. It is a sterol, which means that cholesterol is a combination of a steroid and an alcohol.



It is an important component of cell membranes and is also the basis for the synthesis of other steroids, including the sex hormones estradiol and testosterone, as well as other steroids such as cortisone and vitamin D. In the cell membrane, the steroid ring structure of cholesterol provides a rigid hydrophobic structure that helps boost the rigidity of the cell membrane.

Without cholesterol the cell membrane would be too fluid. In the human body, cholesterol is synthesized in the liver.

Structure of cholesterol

Cholesterol is insoluble in the blood, so when it is released into the blood stream it forms complexes with lipoproteins. Cholesterol can bind to two types of lipoprotein, called high-density lipoprotein (HDL) and low-density lipoprotein (LDL). A lipoprotein is a spherical molecule with water soluble proteins on the exterior. Therefore, when cholesterol is bound to a lipoprotein, it becomes blood soluble and can be transported throughout the body. HDL cholesterol is transported back to the liver. If HDL levels are low, then the blood level of cholesterol will increase. High levels of blood cholesterol are associated with plaque formation in the arteries, which can lead to heart disease and stroke. While most cholesterol in the body is synthesized in the liver, dietary cholesterol also adds to the total blood levels. Cholesterol intake from the diet enters the bloodstream in the LDL form. This helps explain why consumption of foods with high-cholesterol content can lead to increased blood levels of cholesterol which is bad for health. So reducing the cholesterol in the diet can lower the blood level of cholesterol. This can reduce the amount of plaque formation. Aerobic exercise also contributes to health by increasing HDL levels in the blood. Hence more cholesterol is returned to the liver leading to a lower blood level of cholesterol, and reduced plaque formation.

VII. Messenger Lipids: Eicosanoids

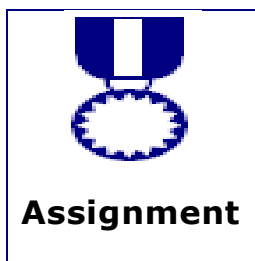
The functions of prostaglandins in physiological processes Essential fatty acids in the cell membranes serve as stores from which prostaglandins are formed. Prostaglandins regulate bodily functions in the heart, kidneys, liver, lungs, brain, nerves and the immune system. They are fatty acid metabolites (derived from arachidonic acid) formed by enzyme cyclooxygenase. They are autacoid hormones, are widely distributed in many tissues, and exert diverse biological effects, including vasodilation, smooth muscle contraction or

relaxation, and regulation of renal function. Prostaglandins are important for the regulation of a host of bodily functions including:

- Inflammation, swelling and pain
- Pressure in the eye, joints or blood vessels
- Secretions from mucus membranes and their viscosity smooth muscle & autonomic reflexes, gastrointestinal, arterial, ear, heart
- Water retention
- Blood clotting ability
- Allergic response
- Rheumatoid arthritis • nerve transmission
- Steroid production & hormone synthesis

In human beings, there are three families of prostaglandins, each of which is derived from a different fatty acid. One prostaglandin, Prostaglandin E1, derived from Gamma linolenic Acid (GLA) and Dihomogamma linolenic Acid (DGLA), is believed to be responsible for controlling several critical functions in the body.

ASSIGNMENT



Detail the clinical conditions associated with nucleic acids and lipid biomolecules